

TE 452 / TE 462 - Telecommunications Policy & Regulation

300 Multiple-Choice Questions with Answers

Part 1

1. A mobile operator lowers the monthly price of its data bundle and observes that the total quantity of bundles subscribers buy rises. Which principle does this most directly illustrate?

- A. The law of supply, where quantity supplied rises as price falls in the short run
- B. The law of demand, where quantity demanded rises as price falls, other things equal ✓**
- C. A rightward shift of the demand curve caused by the price reduction
- D. An increase in demand elasticity that raises the operator's total revenue

Answer: B. The law of demand, where quantity demanded rises as price falls, other things equal

2. An analyst notes that the quantity of broadband demanded changed solely because the price of broadband itself changed, with incomes and tastes unchanged. This is best described as:

- A. A rightward shift of the entire demand curve
- B. A movement along the existing demand curve ✓**
- C. A leftward shift of the entire demand curve
- D. A change in the position of the supply curve

Answer: B. A movement along the existing demand curve

3. When the price of fresh yoghurt rises sharply, many buyers switch to alternatives such as Fanyogo or Fanice instead. In demand analysis these alternatives are best classified as:

- A. Complementary goods
- B. Substitute goods ✓**
- C. Inferior goods
- D. Giffen goods

Answer: B. Substitute goods

4. Among the many determinants of the quantity demanded of a good, economic theory identifies one variable as central to the basic demand relationship. That variable is the:

- A. Number of sellers operating in the market
- B. Cost of the inputs used in production
- C. Level of technology available to producers
- D. Price of the good itself ✓**

Answer: D. Price of the good itself

5. At higher prices a telecom equipment vendor finds it profitable to offer more units, while at lower prices it offers fewer. This relationship between price and quantity offered is captured by the:

- A. Law of demand
- B. Law of supply ✓**
- C. Law of diminishing marginal utility
- D. Principle of tax incidence

Answer: B. Law of supply

6. The market supply curve is drawn holding input costs constant. If the price of a key input such as sugar falls for ice-cream makers, the most likely result is that the supply curve:

- A. Shifts to the left, as producers supply less at each price
- B. Becomes perfectly vertical at the equilibrium quantity
- C. Stays fixed while the quantity demanded increases along it

D. Shifts to the right, as producers supply more at each price ✓

Answer: D. Shifts to the right, as producers supply more at each price

7. Demand for a particular good is described as elastic. This means that:

A. Quantity demanded responds substantially to a change in the good's price ✓

B. Quantity demanded barely responds to a change in the good's price

C. Quantity supplied responds substantially to a change in the good's price

D. Price remains fixed regardless of changes in quantity demanded

Answer: A. Quantity demanded responds substantially to a change in the good's price

8. A regulator studies a service that consumers treat as a necessity with few substitutes, so a price increase changes the quantity bought only slightly. Demand for this service is best described as:

A. Elastic

B. Inelastic ✓

C. Perfectly elastic

D. Unit elastic

Answer: B. Inelastic

9. On a standard supply-and-demand diagram, the single point at which the supply curve and the demand curve intersect determines the:

A. Price ceiling and the resulting shortage

B. Price floor and the resulting surplus

C. Tax incidence borne by buyers and sellers

D. Equilibrium price and equilibrium quantity ✓

Answer: D. Equilibrium price and equilibrium quantity

10. To keep a service affordable, a government sets a maximum legal price below the market equilibrium price. Standard analysis predicts this price ceiling will most likely create:

A. A surplus, because quantity supplied exceeds quantity demanded

B. A shortage, because quantity demanded exceeds quantity supplied ✓

C. A new stable equilibrium exactly at the controlled price

D. An increase in supply that quickly eliminates excess demand

Answer: B. A shortage, because quantity demanded exceeds quantity supplied

11. A government sets a minimum legal price above the market equilibrium to protect suppliers. The most likely consequence of this price floor is:

A. A shortage, because quantity demanded exceeds quantity supplied

B. An immediate return of the market to its equilibrium

C. A reduction in quantity supplied below the quantity demanded

D. A surplus, because quantity supplied exceeds quantity demanded ✓

Answer: D. A surplus, because quantity supplied exceeds quantity demanded

12. In public finance, the term tax incidence refers specifically to:

A. The statutory tax rate that a government formally sets out in legislation

B. The total amount of revenue that a government collects from levying the tax

C. How the burden of a tax is distributed among the participants in a market ✓

D. The administrative cost a government incurs in collecting and enforcing the tax

Answer: C. How the burden of a tax is distributed among the participants in a market

13. When a per-unit tax is imposed on the sellers of a good, the standard model represents its initial effect as:

A. A rightward shift of the supply curve

B. A leftward shift of the demand curve

C. A leftward (upward) shift of the supply curve ✓

D. A rotation that makes the demand curve more elastic

Answer: C. A leftward (upward) shift of the supply curve

14. Two markets face an identical per-unit tax. In Market A demand is highly inelastic; in Market B demand is highly elastic. Compared with sellers, buyers will tend to bear the larger share of the tax in:

- A. Market B, where demand is elastic
- B. Market A, where demand is inelastic ✓**
- C. Both markets equally, regardless of elasticity
- D. Neither market, because the legal duty to remit the tax falls on sellers

Answer: B. Market A, where demand is inelastic

15. An economic system in which production and allocation follow long-standing customary patterns and local leaders have the ultimate say is best described as a:

- A. Command economy, directed by a central government authority
- B. Market economy, coordinated primarily through prices and competition
- C. Traditional economy, based on inherited customs and social roles ✓**
- D. Mixed economy, blending market forces with state intervention

Answer: C. Traditional economy, based on inherited customs and social roles

16. Twentieth-century experience suggests that a command economy tends to fail largely because:

- A. Prices adjust too quickly for state producers to keep pace with the rapidly shifting patterns of consumer demand
- B. Private firms refuse to compete once the state withdraws support and guaranteed contracts
- C. A central authority cannot efficiently gather and act on the information needed to allocate resources ✓**
- D. Consumers are given so many product choices that planners cannot coordinate production

Answer: C. A central authority cannot efficiently gather and act on the information needed to allocate resources

17. A mixed economy is best characterised as one in which:

- A. A central planner sets most prices, wages and output levels across the whole economy
- B. Custom and tradition determine most economic decisions, with little role for market pricing
- C. Market forces operate freely with minimal government involvement in the economy
- D. Markets allocate most resources while governments intervene to correct market failures ✓**

Answer: D. Markets allocate most resources while governments intervene to correct market failures

18. Advocates argue that a market economy can be self-optimising because, when individuals are free to pursue their own interests, the system tends to:

- A. Eliminate the need for formal property rights, contracts or enforcement mechanisms
- B. Produce an equal distribution of income across the whole of society
- C. Prevent monopolies from arising in the economy's key industries
- D. Settle at a stable equilibrium of supply, consumption and prices ✓**

Answer: D. Settle at a stable equilibrium of supply, consumption and prices

19. In a perfectly competitive market each individual buyer and seller is assumed to be a price taker. This assumption follows from the fact that:

- A. A single dominant firm sets the price that other participants then follow
- B. The government fixes the price at which trades in the market must occur
- C. There are so many participants that no single one can influence the market price ✓**
- D. Buyers and sellers openly collude with one another to maintain a single common price

Answer: C. There are so many participants that no single one can influence the market price

20. Compared with a competitive industry, a profit-maximising monopoly is generally expected to:

- A. Produce more output and charge a lower price
- B. Produce less output and charge a higher price ✓**

- C. Produce the same output but at a lower cost
- D. Charge a lower price while expanding total output

Answer: B. Produce less output and charge a higher price

21. An industry is termed a natural monopoly when:

- A. Two or more competing firms tend to supply the market more cheaply than one
- B. The government bars rival firms and licenses a single one to serve the market
- C. Firms in the industry naturally collude to behave as though they were one
- D. A single firm can supply the whole market at lower cost than two or more firms could ✓**

Answer: D. A single firm can supply the whole market at lower cost than two or more firms could

22. Where a genuine natural monopoly exists, a government wishing to avoid wasteful duplication of infrastructure may reasonably choose to:

- A. Tolerate the single firm but subject it to regulation ✓**
- B. Force several firms to compete regardless of the cost of duplication
- C. Withdraw from oversight and leave the market unregulated
- D. Nationalise competing firms throughout the wider economy

Answer: A. Tolerate the single firm but subject it to regulation

23. Within the logic of a market economy, the primary justification for governments intervening through regulators and other institutions is that:

- A. Markets can fail, and intervention may correct such failures ✓**
- B. Markets tend to allocate resources perfectly without outside help
- C. Government planning is inherently superior to market coordination
- D. Competition should be eliminated to simplify oversight

Answer: A. Markets can fail, and intervention may correct such failures

24. An economist distinguishes a change in demand from a change in quantity demanded. A change in demand (a shift of the curve) is caused by:

- A. A change in the good's own price alone, holding other factors constant
- B. A movement to a different point along the same, unchanged demand curve
- C. A change in a determinant other than the good's own price, such as income or tastes ✓**
- D. A simultaneous change in market supply that happens to offset the observed shift in demand exactly

Answer: C. A change in a determinant other than the good's own price, such as income or tastes

25. One reason a monopoly is considered socially harmful is that its output is typically produced:

- A. More efficiently and at a lower cost than under competition
- B. Less efficiently and at a higher cost than under competition ✓**
- C. At exactly the competitive level of cost and efficiency
- D. Chiefly when it is directly subsidised by the government

Answer: B. Less efficiently and at a higher cost than under competition

26. Before the 1980s, telecommunications services in most countries were typically provided by:

- A. Numerous competing private operators
- B. Independent regional cooperatives
- C. State-owned monopolies ✓**
- D. Foreign-owned multinational consortia

Answer: C. State-owned monopolies

27. A factor underpinning the pre-1980s telecom monopoly paradigm was economic: high network build costs and low demand meant a second operator could not easily recover its costs. This gave the industry the character of a:

- A. Natural monopoly ✓**

- B. Perfectly contestable market
- C. Fully deregulated free market
- D. Command-economy consumer good

Answer: A. Natural monopoly

28. A political factor behind early state telecom monopolies was that governments:

- A. Were discouraged by international postal treaty from licensing a private telecom firm
- B. Lacked a government ministry with the technical capacity to run even a basic postal service
- C. Preferred to auction off radio-spectrum rights directly to the highest-bidding private investor of the day

D. Wished to retain state control over communications, often treating them as an extension of postal networks ✓

Answer: D. Wished to retain state control over communications, often treating them as an extension of postal networks

29. In telecommunications, liberalization is best defined as:

- A. The transfer of an operator's ownership from the state to private hands
- B. The intervention of government to achieve specific market outcomes
- C. The process of transition from a monopoly market to a competitive one ✓**
- D. The setting of technical standards shared across the operators

Answer: C. The process of transition from a monopoly market to a competitive one

30. A central element of liberalization is the removal of exclusivities from the former monopoly fixed-network operator, commonly called the incumbent. The direct effect of removing these exclusivities is to:

- A. Allow new entrants to compete in services that were previously monopolised ✓**
- B. Transfer ownership of the incumbent's shares to a group of private investors
- C. Secure the incumbent a long, protected and exclusive market position
- D. Impose strict price caps on each retail service the incumbent operator offers

Answer: A. Allow new entrants to compete in services that were previously monopolised

31. Privatization in the telecom sector refers to:

- A. The transition of an entire market from monopoly toward open competition
- B. The imposition of universal-service obligations on licensed operators
- C. The transfer of ownership of an operator from the state to the private sector ✓**
- D. The creation of a newly independent national regulatory authority

Answer: C. The transfer of ownership of an operator from the state to the private sector

32. Governments have privatized incumbent operators in different ways. Selling a stake to a single strategic investor, rather than via a broad public share sale, has been the more common approach in:

- A. Highly industrialised countries
- B. Countries that have no incumbent operator
- C. Countries that retained a full state monopoly
- D. Developing countries ✓**

Answer: D. Developing countries

33. A country opens its mobile market to several competing firms but keeps the incumbent fixed-line operator fully state-owned. This situation is best described as:

- A. Privatization of the incumbent operator without opening the market to competition
- B. Liberalization of the market without privatization of the incumbent operator ✓**
- C. Both full market liberalization and full privatization of the incumbent
- D. Neither liberalization of the market nor privatization of the incumbent operator

Answer: B. Liberalization of the market without privatization of the incumbent operator

34. In the context of the telecom sector, regulation is best defined as:

- A. Government intervention in markets to achieve certain desired outcomes ✓**

- B. The complete withdrawal of government from market activity
- C. The wholesale transfer of state-owned assets into full private ownership
- D. The voluntary self-policing of firms with absolutely no external oversight

Answer: A. Government intervention in markets to achieve certain desired outcomes

35. Regulation is sometimes said to take three basic forms. The form that establishes a common, legally enforceable platform for all businesses, such as rules governing electrical supply or company accounting, is:

- A. Competition regulation
- B. Public policy regulation
- C. Technical standardization ✓**
- D. Voluntary self-regulation

Answer: C. Technical standardization

36. Legislation that controls the supply of goods such as alcohol, gambling or pornography because doing so is believed to serve the public interest is an example of:

- A. Technical standardization
- B. Competition regulation
- C. Public policy regulation ✓**
- D. Co-regulation

Answer: C. Public policy regulation

37. The form of regulation that seeks to create and maintain fair competitive relationships between businesses, and between businesses and consumers, is:

- A. Technical standardization
- B. Competition regulation ✓**
- C. Public policy regulation
- D. Zero regulation

Answer: B. Competition regulation

38. Although telecom regulation can take all three basic forms, the principal focus of contemporary regulation of the telecommunications services industry is:

- A. The standardisation of accounting rules across many industries
- B. The control of socially sensitive goods such as gambling
- C. The creation, nourishment and maintenance of competitive markets ✓**
- D. The elimination of private ownership from the sector

Answer: C. The creation, nourishment and maintenance of competitive markets

39. Among the widely accepted regulatory objectives in the telecom sector is the aim to:

- A. Secure a fixed profit margin for each licensed operator
- B. Restrict connectivity to commercially attractive urban areas
- C. Promote universal access to basic telecommunications services ✓**
- D. Prevent new operators from entering the market

Answer: C. Promote universal access to basic telecommunications services

40. A recognised regulatory objective is that, where competitive markets do not exist or have failed, the regulator should:

- A. Shield the dominant firm from potential competitors in the market
- B. Prevent abuses of market power such as excessive pricing by dominant firms ✓**
- C. Withdraw regulatory oversight and let the market correct itself unaided
- D. Fix operators' retail prices at exactly identical levels nationwide

Answer: B. Prevent abuses of market power such as excessive pricing by dominant firms

41. Which statement best captures the distinction between policy and law as drawn in the course material?

- A. Law can compel or prohibit behaviour, whereas policy merely guides actions toward a**

desired outcome ✓

- B. Policy can compel or prohibit behaviour, whereas law offers non-binding guidance
- C. Policy and law are identical concepts, so the two terms may be used interchangeably
- D. Regulation sets the broad long-term goals while policy enforces those goals through the national courts

Answer: A. Law can compel or prohibit behaviour, whereas policy merely guides actions toward a desired outcome

42. In the policy-to-practice chain, regulation is primarily concerned with:

- A. Translating the objectives of policy decisions into practical rules and decisions ✓**
- B. Setting the broad, long-term national vision that exists prior to policy
- C. Replacing the need for underlying legislation or statutory authority
- D. Auditing the personal finances of individual government ministers each year

Answer: A. Translating the objectives of policy decisions into practical rules and decisions

43. In many countries before reform, a single ministry or government unit often acted simultaneously as:

- A. An independent regulator that is fully separated from the operator
- B. A commercial firm with no policy responsibilities of its own
- C. Policy maker, policy implementer and operator of the telephone service ✓**
- D. A neutral standards body with no operational role in the sector

Answer: C. Policy maker, policy implementer and operator of the telephone service

44. The International Telecommunication Union (ITU) is best described as:

- A. A private consortium formed and funded by the world's largest mobile network operators worldwide
- B. A regional trade bloc that negotiates import tariffs on telecommunications equipment for its members
- C. A specialised agency of the United Nations responsible for information and communication technology issues ✓**
- D. A commercial standards-setting firm that is owned and controlled by major equipment vendors

Answer: C. A specialised agency of the United Nations responsible for information and communication technology issues

45. Within the ITU's three-sector structure, the sector that manages the international radio-frequency spectrum and satellite-orbit resources is:

- A. ITU-R (Radiocommunication), one of the ITU's three constituent sectors ✓**
- B. ITU-T (Standardization), one of the ITU's three constituent sectors
- C. ITU-D (Development), one of the ITU's three constituent sectors
- D. ITU-P (Policy), one of the ITU's three constituent sectors

Answer: A. ITU-R (Radiocommunication), one of the ITU's three constituent sectors

46. The ITU sector responsible for standardizing global telecommunications, except for radio matters, is:

- A. ITU-R (Radiocommunication)
- B. ITU-D (Development)
- C. ITU-T (Standardization) ✓**
- D. ITU-S (Spectrum)

Answer: C. ITU-T (Standardization)

47. Helping to spread equitable, sustainable and affordable access to ICTs is the principal role of which ITU sector?

- A. ITU-R (Radiocommunication)
- B. ITU-D (Development) ✓**
- C. ITU-T (Standardization)
- D. ITU-A (Access)

Answer: B. ITU-D (Development)

48. The World Trade Organization's Agreement on Basic Telecommunications (the BTA) primarily:

- A. Allocates radio spectrum directly to individual telecom operators in each member state
- B. Establishes rules for competition and promotes open investment in telecoms across countries ✓**
- C. Sets the maximum retail tariffs that operators may charge their consumers directly
- D. Issues operating licences directly to individual carriers within each member state

Answer: B. Establishes rules for competition and promotes open investment in telecoms across countries

49. One of the six basic principles of the WTO Reference Paper on Regulation is that:

- A. The dominant operator should also serve as the regulator
- B. Licensing criteria should be kept confidential from competitors
- C. Scarce resources should be reserved for the incumbent operator
- D. There should be a regulatory body independent of operating companies ✓**

Answer: D. There should be a regulatory body independent of operating companies

50. The WTO Reference Paper requires telecom regulators to be independent of operating companies. On independence from government, the material indicates that regulators are:

- A. Not necessarily independent of governments, though greater independence is often regarded as better ✓**
- B. Required by law to be controlled and directed by the relevant government ministry
- C. Required by law to be formally independent of government in most respects
- D. Required by law to be jointly owned and controlled by the very operators that they are meant to regulate

Answer: A. Not necessarily independent of governments, though greater independence is often regarded as better

51. According to the framework for regulation, the laws passed by governments to enable regulation typically:

- A. Secure long-term monopoly status for the incumbent state operator
- B. Appoint regulatory bodies and confer on them powers to take action ✓**
- C. Remove accountability and reporting requirements from regulators
- D. Fix the retail prices of each regulated service nationwide

Answer: B. Appoint regulatory bodies and confer on them powers to take action

52. Which option best lists the categories of law that bear specifically on the telecommunications services industry?

- A. Sector-specific telecom law, competition and fair-trading law, and other relevant law such as privacy ✓**
- B. Criminal law, international maritime law and constitutional law relating to the state
- C. Tax law, general employment law and intellectual-property law covering ordinary commercial dealings
- D. Banking law, insurance law and environmental law drawn from other regulated sectors

Answer: A. Sector-specific telecom law, competition and fair-trading law, and other relevant law such as privacy

53. Provisions enacted by bodies such as ECOWAS or the WTO take force in a member or signatory country when they are:

- A. Transposed into national law in accordance with treaty obligations ✓**
- B. Simply published on the regional body's official website
- C. Approved by a majority vote of the country's licensed operators
- D. Adopted the moment they are agreed upon at the regional level

Answer: A. Transposed into national law in accordance with treaty obligations

54. The framework notes that the function of regulation may be split between two or more bodies. A risk a country must manage when doing so is that:

A. Poorly designed division of functions can lead to disagreements and slow decision-making ✓

B. A single national regulator will inevitably end up capturing the entire market

C. Operators may find it much harder to obtain a licence from the divided bodies

D. Spectrum ends up allocated to operators free of charge under the divided arrangement

Answer: A. Poorly designed division of functions can lead to disagreements and slow decision-making

55. Listed in increasing order of intrusiveness, the styles of regulation are:

A. Formal regulation, co-regulation, self-regulation, zero regulation

B. Zero regulation, self-regulation, co-regulation, formal regulation ✓

C. Self-regulation, zero regulation, formal regulation, co-regulation

D. Co-regulation, formal regulation, zero regulation, self-regulation

Answer: B. Zero regulation, self-regulation, co-regulation, formal regulation

56. An industry described as operating under zero regulation is one in which:

A. A regulator intervenes minutely in the firm's daily operations

B. Firms jointly agree binding rules under an industry watchdog

C. Prices are set directly by a government ministry

D. Market forces are allowed to work with no external intervention ✓

Answer: D. Market forces are allowed to work with no external intervention

57. Self-regulation, in which the players in an industry regulate themselves, tends to be MOST effective when:

A. There is a high convergence of interest among the stakeholders ✓

B. There are strong commercial conflicts between the players

C. A single firm dominates and distrusts the other players

D. The regulator intervenes in routine operational decisions

Answer: A. There is a high convergence of interest among the stakeholders

58. A frequently cited benefit of self-regulation is that:

A. Companies usually understand their own businesses better than an external regulator does ✓

B. It removes the administrative costs of external regulation from both consumers and the wider industry

C. It ensures that companies rarely act opportunistically toward their rivals

D. It eliminates the need for underlying legislation or a licensing regime

Answer: A. Companies usually understand their own businesses better than an external regulator does

59. Co-regulation is best described as an arrangement in which:

A. Firms operate independently with little external oversight of their conduct

B. The regulator alone dictates the technical and pricing decisions of the firms unilaterally from outside

C. The regulator joins industry players in a joint forum rather than regulating purely from outside ✓

D. A foreign international body simply assumes the domestic regulatory authority

Answer: C. The regulator joins industry players in a joint forum rather than regulating purely from outside

60. While potentially very successful, co-regulation is particularly vulnerable to:

A. An immediate and total loss of the regulator's formal powers

B. The automatic capture of the regulator by consumer groups

C. Role ambiguity and differing perceptions of what each player will do ✓

D. A blanket legal prohibition on industry collaboration of this kind

Answer: C. Role ambiguity and differing perceptions of what each player will do

61. Regulation is described as intrusive when the regulator:

- A. Lays down general principles and leaves the methods to firms themselves
- B. Withdraws from the market and lets market forces operate unaided
- C. Intervenes often and minutely in the day-to-day running of a company's business ✓**
- D. Limits itself to publishing occasional, non-binding guidelines

Answer: C. Intervenes often and minutely in the day-to-day running of a company's business

62. One reason intrusive regulation is considered undesirable is that:

- A. It tends to sharply reduce the total administrative cost of closely regulating the whole industry
- B. It assumes that competitive market forces will prevail without oversight
- C. The regulator effectively becomes a proxy manager of the industry, weakening operators' freedom ✓**
- D. It shifts the regulator's responsibility for market outcomes onto the firms

Answer: C. The regulator effectively becomes a proxy manager of the industry, weakening operators' freedom

63. A good regulator is advised to make regulation pervasive rather than intrusive. This means regulation should:

- A. Influence operators broadly while letting them manage their own affairs within the spirit of the rules ✓**
- B. Dictate the exact, precise technical solution required of each licensed operator
- C. Have as little practical impact as possible on the day-to-day affairs of companies operating in the market
- D. Apply narrowly to a single dominant operator, leaving the other firms untouched

Answer: A. Influence operators broadly while letting them manage their own affairs within the spirit of the rules

64. The main instruments through which regulation is accomplished include authorizations, rules, determinations, consents and:

- A. Guidelines ✓**
- B. Share buy-backs
- C. Spectrum auctions
- D. Corporate tax assessments

Answer: A. Guidelines

65. A system of authorization may issue specific authorizations and general authorizations. These correspond respectively to:

- A. Spectrum licences and national numbering plans
- B. Individual operator licences and class licences ✓**
- C. Regulatory price caps and interconnection orders
- D. Formal consents and regulatory determinations

Answer: B. Individual operator licences and class licences

66. Determinations are instruments of ongoing regulation that, in effect:

- A. Revoke an operator's licence outright, leaving no right of appeal against the regulator's decision
- B. Replace the underlying legislation abruptly and without warning
- C. Elaborate the rules, for example in the periodic review of price controls ✓**
- D. Carry no legal force and are merely advisory in nature

Answer: C. Elaborate the rules, for example in the periodic review of price controls

67. A consent is best understood as:

- A. An order that formally imposes an additional new obligation on the operator concerned
- B. An informal guideline document that carries no legal status whatsoever in practice
- C. A formal financial penalty levied against an operator for breach of a licence condition
- D. A negative determination that formally relieves an operator from complying with a requirement ✓**

Answer: D. A negative determination that formally relieves an operator from complying with a requirement

68. Guidelines differ from rules and determinations in that guidelines:

A. Have far greater binding legal force than the licence conditions attached to the operator do themselves

B. Do not in themselves possess legal status and are usually written in user-friendly language ✓

C. Can be issued by the courts, following a successful appeal process

D. Revoke a licence the very moment they are breached by the operator

Answer: B. Do not in themselves possess legal status and are usually written in user-friendly language

69. A key reason regulators consult widely before finalising instruments is that consultation:

A. Transfers the formal decision-making power directly to the consulted parties

B. Removes the regulator's legal obligation to justify its decisions

C. Gives industry ownership and pre-knowledge of decisions and enriches decision-making ✓

D. Ensures that consulted parties rarely appeal the final regulatory outcome

Answer: C. Gives industry ownership and pre-knowledge of decisions and enriches decision-making

70. An appeal against a regulator's action may be brought on the grounds of law (illegality), procedure (misadministration) or:

A. Profitability (commercial loss)

B. Logic (irrationality) ✓

C. Popularity (public disapproval)

D. Convenience (administrative delay)

Answer: B. Logic (irrationality)

71. Which of the following is a recognised enforcement power available to a regulator dealing with rule breaches?

A. Seizing ownership of a competitor's shares

B. Setting the personal salaries of company directors

C. Revoking a licence, or imposing fines and compliance orders ✓

D. Banning the company's customers from switching providers

Answer: C. Revoking a licence, or imposing fines and compliance orders

72. Regarding the funding of national regulatory authorities, the typical arrangement is that:

A. Operators are legally barred from contributing funds to the regulator's budget

B. Regulators are funded mainly by the fines they collect from ordinary telecom consumers each year

C. Funding comes mainly from international donor agencies and foreign aid grants

D. Governments assume primary responsibility but usually pass the cost onto industry through fees ✓

Answer: D. Governments assume primary responsibility but usually pass the cost onto industry through fees

73. The concept of regulatory hazard refers to the fact that:

A. Regulators personally face genuine physical danger when inspecting live telecom networks

B. Regulation removes most of the commercial risk from the telecom market

C. Operators face absolutely no consequences whatsoever for repeatedly breaching the rules

D. Regulatory power, if exercised unwisely, can distort markets and damage the industry it serves ✓

Answer: D. Regulatory power, if exercised unwisely, can distort markets and damage the industry it serves

74. Regulatory capture occurs when:

A. The regulator becomes paralysed and stops issuing decisions for a long period

B. Individual operators are captured and then run directly by the regulator itself acting as their owner

C. Ordinary consumers themselves take over the entire day-to-day running of the regulator

D. The regulator identifies too closely with, and so serves the interests of, the industry it regulates ✓

Answer: D. The regulator identifies too closely with, and so serves the interests of, the industry it regulates

75. The information asymmetry, or knowledge gap, that regulators face arises because:

- A. Regulators are legally forbidden from collecting the operators' detailed operational and cost data
- B. Consumers deliberately conceal their true preferences from the regulator
- C. Operators are legally required to withhold their internal financial records
- D. The companies being regulated usually know far more about the business than the regulator does ✓**

Answer: D. The companies being regulated usually know far more about the business than the regulator does

76. Regulators issue licences to companies primarily to authorise them to:

- A. Operate telecommunications networks and provide telecommunications services ✓**
- B. Manufacture and export telecommunications equipment internationally
- C. Set the national monetary and fiscal policy for the entire economy
- D. Audit the financial accounts of rival operators each year

Answer: A. Operate telecommunications networks and provide telecommunications services

77. Beyond granting permission to operate, a telecommunications licence characteristically specifies:

- A. The exact retail prices that other licensed operators in the country must charge their retail consumers
- B. The complete internal staffing structure and organisational chart of the regulator itself
- C. What facilities a company may deploy, what services it may offer, and how it must interact with other businesses ✓**
- D. The personal income tax obligations of each of the company's individual board directors

Answer: C. What facilities a company may deploy, what services it may offer, and how it must interact with other businesses

78. In addition to facilities and services, major licences may govern access to scarce resources such as spectrum and numbers, and commonly require companies to pay:

- A. Upfront fees when the licence is awarded and annual payments thereafter ✓**
- B. A fixed share of annual profits paid to competing operators
- C. Nothing, since licences are typically granted free of charge
- D. A single one-time refundable deposit that is fully returned to the operator upon licence expiry

Answer: A. Upfront fees when the licence is awarded and annual payments thereafter

79. Historically, telecom licences were allocated separately for different sub-markets, for example fixed networks, mobile networks or paging. The main characteristic of this approach was that licences:

- A. Were broadly technology- and service-neutral by design
- B. Imposed identical licence terms across each market segment
- C. Differentiated between specific technologies and services ✓**
- D. Were rarely tied to a particular type of technology or service

Answer: C. Differentiated between specific technologies and services

80. Recent technological convergence, which lets different platforms deliver a wider range of services, has prompted some regulators to move toward:

- A. Stricter technology-specific licensing for each platform
- B. Technology- and service-neutral licensing ✓**
- C. The complete abolition of the licensing regime
- D. Permanent exclusive licences for incumbent operators

Answer: B. Technology- and service-neutral licensing

81. A technology-neutral licence to provide a telephone service is one that:

- A. Requires the company to use one single, government-specified fixed-line access technology

- B. Strictly forbids the company from offering voice services of its own
- C. Mandates the use of a single, government-approved equipment vendor
- D. Lets the company choose whether to use a fixed network or a particular wireless technology ✓**

Answer: D. Lets the company choose whether to use a fixed network or a particular wireless technology

82. The two main types of licence awarded to telecom companies are:

- A. Spectrum-specific licences and separate telephone numbering licences
- B. International import licences and international export licences
- C. Short-term provisional licences and longer-term probationary licences
- D. Individual operator licences and class licences (general authorizations) ✓**

Answer: D. Individual operator licences and class licences (general authorizations)

83. Individual operator licences are used primarily where:

- A. The market segment has essentially unlimited room for many new entrants to join
- B. No scarce resources of a technical kind are involved in that market segment
- C. Perfectly identical licence terms are wanted for each business operating in that particular market segment
- D. The number of licences is limited, dominance is likely, or specific resources such as spectrum must be allocated ✓**

Answer: D. The number of licences is limited, dominance is likely, or specific resources such as spectrum must be allocated

Answer: D. The number of licences is limited, dominance is likely, or specific resources such as spectrum must be allocated

84. Class licences (general authorizations) are characterised by the fact that they:

- A. Are tailored individually to just one single, specifically named operator in the market segment
- B. Set out identical terms of operation for all businesses in a market segment, such as ISPs ✓**
- C. Are reserved for use by the former state incumbent operator
- D. Apply in the specific cases where radio spectrum must be allocated

Answer: B. Set out identical terms of operation for all businesses in a market segment, such as ISPs

85. A current trend favours replacing individual operator licences with general authorizations wherever possible because doing so:

- A. Simplifies regulation, helps ensure a level playing field, and is more flexible to market change ✓**
- B. Maximises the one-time upfront revenue collected by the state from each individual licensed operator
- C. Locks in the incumbent's fully protected monopoly position for the long term
- D. Removes the need for an independent regulator in the sector

Answer: A. Simplifies regulation, helps ensure a level playing field, and is more flexible to market change

86. One licensing objective treats basic telecommunications as an essential public service. This justifies:

- A. Removing regulatory oversight the moment the market is privatised
- B. Retaining some controls to ensure services are provided in the public interest ✓**
- C. Reserving the provision of service for a single state-owned monopoly operator indefinitely
- D. Allowing operators to serve just the most profitable customers

Answer: B. Retaining some controls to ensure services are provided in the public interest

87. Network roll-out and service-coverage obligations are often written into licences to advance which licensing objective?

- A. Generation of one-time government spectrum auction revenue
- B. Protection of the incumbent operator from new competition
- C. Standardisation of company accounting and reporting practices
- D. Expansion of networks and services, and universal service ✓**

Answer: D. Expansion of networks and services, and universal service

88. When a state-owned incumbent is privatized, a licence is necessary because it:

- A. Transfers full legal ownership and operational control of the regulator to the investor
- B. Specifies what the investor is buying and what the government expects from the operator ✓**
- C. Assures the investor lasting freedom from future regulation
- D. Removes the operator's obligation to serve less profitable customers

Answer: B. Specifies what the investor is buying and what the government expects from the operator

89. Licensing can be used to regulate market structure, most directly by:

- A. Fixing the wholesale price of each network input
- B. Determining the number of operators licensed to provide services ✓**
- C. Setting the wages paid to an operator's employees
- D. Selecting which equipment vendors the licensed operators are required to use

Answer: B. Determining the number of operators licensed to provide services

90. Licence conditions intended to create a level playing field and limit incumbents' abuse of dominance are generally referred to as:

- A. Universal-service obligations for underserved regions
- B. Spectrum-refarming and reallocation provisions
- C. Tariff-rebalancing and price-adjustment schedules
- D. Anti-competitive safeguards or fair-trading conditions ✓**

Answer: D. Anti-competitive safeguards or fair-trading conditions

91. Finite resources such as radio spectrum, numbers and rights of way should, as a licensing objective, be allocated:

- A. In full to whichever operator has the longest operating history
- B. On a first-come basis with little regard to efficiency of use
- C. To the operators that are owned by the national government
- D. Between operators fairly, efficiently and in the public interest ✓**

Answer: D. Between operators fairly, efficiently and in the public interest

92. Licensing can generate government revenue in two characteristic ways:

- A. One-time revenues from auctions and continuing revenue from annual licence fees ✓**
- B. Confiscation of operator profits and the outright seizure of their network assets by the state
- C. Personal income taxes levied directly on the regulator's own staff
- D. Mandatory donations collected annually from international agencies

Answer: A. One-time revenues from auctions and continuing revenue from annual licence fees

93. Consumer-protection conditions in telecom licences commonly address matters such as:

- A. Billing practices, complaint mechanisms and mandatory services like emergency calls ✓**
- B. The detailed internal share structure of the operator's holding company
- C. The personal credit ratings of each of the company's individual board directors
- D. The choice of which external auditors are used by the regulator itself

Answer: A. Billing practices, complaint mechanisms and mandatory services like emergency calls

94. By clearly defining the rights and obligations of the operator and the regulator, a licence advances the objective of:

- A. Regulatory certainty, which builds confidence and helps attract investment ✓**
- B. Spectrum refarming carried out across competing operators
- C. Eliminating the regulator's discretion over its own decisions
- D. Assuring each operator an exactly equal share of the national market for services

Answer: A. Regulatory certainty, which builds confidence and helps attract investment

95. Radio spectrum is a scarce resource whose international allocation is governed by rules intended to maximise its use while minimising cross-border interference. These rules are managed by the:

- A. International Telecommunication Union (ITU) ✓**
- B. World Trade Organization (WTO)
- C. Economic Community of West African States (ECOWAS)
- D. Individual operators within each country

Answer: A. International Telecommunication Union (ITU)

96. Although telephone numbers are in principle infinite, they are treated as a scarce resource partly because:

- A. There are physically just a very small, fixed number of usable telephone-number digits in existence across the world
- B. Each individual telephone number requires its own dedicated radio frequency to function
- C. A given telephone number can be assigned to and used by just one single country worldwide
- D. People can hold only a limited number of digits in short-term memory, and numbers must be allocated in meaningful blocks ✓**

Answer: D. People can hold only a limited number of digits in short-term memory, and numbers must be allocated in meaningful blocks

97. Number portability is a mechanism by which:

- A. The operator retains permanent ownership of the telephone numbers that it issues to its subscribers
- B. The subscriber, rather than the operator, owns the number and can take it when switching providers ✓**
- C. Numbers are periodically auctioned off to whichever operator happens to bid the highest
- D. Subscribers must change their number each time they switch to a new provider

Answer: B. The subscriber, rather than the operator, owns the number and can take it when switching providers

98. In allocating spectrum, a regulator may auction it to the highest bidder or allocate it at low cost. Allocating spectrum at low cost is typically intended to:

- A. Reduce prices or encourage the rollout of new services ✓**
- B. Maximise the one-time revenue collected by the government
- C. Reserve the band for use by a single operator
- D. Remove the need to manage interference between users

Answer: A. Reduce prices or encourage the rollout of new services

99. A primary reason spectrum must be allocated through agreements such as licences, rather than used freely, is to:

- A. Increase the total number of digits available for telephone numbers nationally
- B. Let each licensed operator broadcast on the frequency it prefers
- C. Remove the need for international coordination of frequencies
- D. Ensure it is properly distributed and that users do not interfere with one another ✓**

Answer: D. Ensure it is properly distributed and that users do not interfere with one another

100. The core principle of interconnection is that:

- A. Each operator should deliberately keep its own subscribers isolated from the networks of its rivals
- B. Operators of a roughly similar size are the ones required to interconnect
- C. Interconnection is legally required between the largest international carriers
- D. Everyone should be able to communicate with everyone else, regardless of which network they subscribe to ✓**

Answer: D. Everyone should be able to communicate with everyone else, regardless of which network they subscribe to

Part 2

1. When the price of a mobile voice call rises by 10 percent and, as a result, the demand for SMS messages increases by 15 percent, the cross-price elasticity of demand between calls and SMS is positive. This indicates the two services are:

- A. Complementary goods — consumers buy more of both when the price of calls rises
B. Substitute goods — consumers switch from calls to SMS as calls become more expensive ✓
C. Independent goods — the demand for SMS is unrelated to the price of calls
D. Inferior goods — the demand for SMS falls as consumer incomes rise
Answer: B. Substitute goods — consumers switch from calls to SMS as calls become more expensive

2. When consumer incomes rise by 10 percent and demand for broadband services rises by 18 percent, the income elasticity of demand for broadband is greater than one. This classifies broadband as:

- A. A luxury (income-elastic) good, whose demand rises proportionally faster than income ✓**
B. A necessity (income-inelastic) good, whose demand barely changes with income
C. An inferior good, whose overall demand actually falls whenever income rises
D. A Giffen good, whose quantity demanded oddly rises as its own price increases
Answer: A. A luxury (income-elastic) good, whose demand rises proportionally faster than income

3. In a competitive telecom market, consumer surplus represents:

- A. The extra profit that a firm earns by charging a price that is well above its own costs
B. The difference between what consumers are willing to pay for a service and what they actually pay ✓
C. The overall reduction in quantity demanded that is caused by a sudden increase in price
D. The specific share of the tax burden that is borne by consumers after a new excise tax
Answer: B. The difference between what consumers are willing to pay for a service and what they actually pay

4. When a monopolist restricts output below the socially efficient level and raises the price, the economic loss to society compared with the competitive outcome is called:

- A. Deadweight loss ✓**
B. Consumer surplus
C. Producer surplus
D. Tax incidence
Answer: A. Deadweight loss

5. A mobile platform operator charges low prices to consumers to attract a large user base, and then charges higher prices to businesses that wish to reach those consumers through advertising or app stores. This business model is characteristic of:

- A. A perfectly competitive market in which numerous small firms are price-takers facing an identical market price
B. A natural monopoly, where one large network serves the whole market at a lower cost than several rivals could
C. A two-sided (or multi-sided) market, where the platform intermediates two distinct customer groups ✓
D. A command economy in which a central authority sets the prices faced by each separate group of consumers
Answer: C. A two-sided (or multi-sided) market, where the platform intermediates two distinct customer groups

6. A social messaging application becomes more valuable to each new user as more people already use it, because more connections are possible. Economists call this phenomenon:

- A. A positive network externality (or network effect) ✓**
B. A negative externality, where each new user imposes costs on existing users
C. Economies of scale, where average costs fall as output rises
D. Price discrimination, where different users are charged different prices
Answer: A. A positive network externality (or network effect)

7. A mobile subscriber who has accumulated loyalty points, contacts stored in a proprietary format, and a memorable number that cannot be easily ported faces high switching costs.

High switching costs benefit the incumbent operator primarily by:

- A. Lowering the average operational cost of serving a subscriber, because loyalty schemes are cheap to run
- B. Strengthening the incentive for new firms to enter, since locked-in customers are seen as easy targets
- C. Reducing the competitive pressure it faces, since subscribers find it costly to move to rivals ✓**
- D. Shifting a share of its existing subscribers across to competing networks as their contracts gradually lapse

Answer: C. Reducing the competitive pressure it faces, since subscribers find it costly to move to rivals

8. A fixed-line operator has laid fibre-optic cable across an entire city. This investment is largely non-recoverable if the operator exits the market. Such costs are termed:

- A. Variable costs, which change with the volume of traffic carried
- B. Opportunity costs, which represent the next-best foregone alternative
- C. External costs, which are imposed on parties outside the transaction
- D. Sunk costs ✓**

Answer: D. Sunk costs

9. A firm whose average cost per subscriber falls steadily as it adds more subscribers is said to exhibit:

- A. Diseconomies of scale, where average cost rises as output increases
- B. Economies of scale ✓**
- C. Constant returns to scale, where average cost does not change with output
- D. Diminishing marginal utility, where each extra unit provides less satisfaction

Answer: B. Economies of scale

10. In a contestable market, even a single firm may price competitively because:

- A. The government fixes prices at the competitive level by decree, leaving the firm little discretion over its tariffs
- B. The incumbent restrains its own profit margins as a gesture of corporate social responsibility toward its users
- C. Sunk costs across the industry are so high that a would-be rival could not recover the fixed capital needed to enter
- D. The threat of potential entry by rivals constrains the incumbent from raising prices above competitive levels ✓**

Answer: D. The threat of potential entry by rivals constrains the incumbent from raising prices above competitive levels

11. A telecommunications firm that charges each customer exactly their maximum willingness to pay, thereby extracting all consumer surplus, is practising:

- A. Second-degree price discrimination, where prices vary by volume consumed
- B. Third-degree price discrimination, where prices vary by customer group
- C. First-degree (or perfect) price discrimination ✓**
- D. Predatory pricing, where prices are set below cost to drive rivals out

Answer: C. First-degree (or perfect) price discrimination

12. A mobile operator offers the same data plan at one price to residential customers and a lower price to students, based on the observation that students are more price-sensitive. This is an example of:

- A. Third-degree price discrimination, segmenting customers by group elasticity ✓**
- B. First-degree price discrimination, where each buyer pays their unique reservation price
- C. Second-degree price discrimination, where prices depend on quantity purchased
- D. Uniform pricing, because the plan content is identical

Answer: A. Third-degree price discrimination, segmenting customers by group elasticity

13. A broadband provider charges subscribers a fixed monthly access fee and then an additional per-gigabyte usage charge on top. This pricing structure is called:

- A. A flat rate, where one fixed monthly fee covers usage regardless of the volume consumed
- B. Predatory pricing, where prices are set below cost to drive rivals from the market
- C. Resale pricing, where the provider buys capacity wholesale and marks it up for retail
- D. A two-part tariff ✓**

Answer: D. A two-part tariff

14. The Lerner index measures market power as (Price minus Marginal Cost) divided by Price. A Lerner index of zero indicates:

A. No market power — the firm is pricing at marginal cost, consistent with perfect competition ✓

- B. Substantial market power — the firm is pricing well above its marginal cost of supply
- C. A natural monopoly — a single efficient firm supplies the whole market at least cost
- D. Government-mandated pricing — the regulator sets the firm's prices directly by decree

Answer: A. No market power — the firm is pricing at marginal cost, consistent with perfect competition

15. Allocative efficiency in a market is achieved when:

- A. A single dominant firm becomes large enough to minimise its own average cost of production across the market
- B. Producers throughout the industry adopt the most technically advanced equipment currently available to them
- C. The government sets and controls the retail price of each separate product traded across the whole economy
- D. Goods and services are produced in the quantities that consumers value them, at prices that reflect true costs ✓**

Answer: D. Goods and services are produced in the quantities that consumers value them, at prices that reflect true costs

16. An operator is productively efficient when it:

- A. Sets its retail prices exactly equal to those charged by its nearest competitor
- B. Allocates its output strictly in proportion to the annual incomes of its individual customers
- C. Earns a rate of return that is exactly equal to the prevailing market rate of interest
- D. Produces its output at the lowest possible cost, given current technology and input prices ✓**

Answer: D. Produces its output at the lowest possible cost, given current technology and input prices

17. A regulatory policy that sacrifices short-term competition concerns in order to encourage investment in next-generation infrastructure is prioritising:

- A. Dynamic efficiency — the ability of the market to innovate and improve over time ✓**
- B. Allocative efficiency — matching quantities produced to consumer preferences
- C. Productive efficiency — minimising the cost of current output
- D. Regulatory capture — the regulator acting in the interest of the regulated firm

Answer: A. Dynamic efficiency — the ability of the market to innovate and improve over time

18. A mobile market with three operators, each aware that its pricing decisions will provoke a response from the other two, is best described as an:

- A. Perfect competition, where no single firm influences price
- B. Oligopoly ✓**
- C. Natural monopoly, where a single firm serves the whole market at least cost
- D. Monopsony, where a single buyer faces many sellers

Answer: B. Oligopoly

19. An incumbent operator temporarily prices a service below its cost of provision with the intention of forcing a new entrant out of the market, after which it plans to raise prices. This behaviour is known as:

- A. Penetration pricing, aimed at legitimately building market share
- B. Marginal cost pricing, required of regulated dominant firms
- C. Predatory pricing ✓**
- D. Price discrimination, charging different buyers different prices

Answer: C. Predatory pricing

20. A telecom operator sells broadband, mobile and television services together at a combined price lower than the sum of their individual prices, requiring customers to subscribe to all three. This commercial strategy is called:

- A. Disaggregated resale, where services are sold separately at wholesale
- B. Co-location, where competitors share the operator's physical infrastructure
- C. Resale, where an operator buys wholesale services and retails them
- D. Bundling ✓**

Answer: D. Bundling

21. Regulators use the Herfindahl-Hirschman Index (HHI) to measure market concentration. An HHI approaching 10,000 indicates:

- A. A perfectly competitive market populated by many small equal-sized firms
- B. A market in which the two largest firms each hold roughly half of the shares
- C. A highly concentrated market approaching a single-firm monopoly ✓**
- D. A market so fragmented that it falls below the threshold for regulatory oversight

Answer: C. A highly concentrated market approaching a single-firm monopoly

22. In mobile markets, spectrum licences granted to a limited number of operators create a barrier to entry because:

- A. Holding a spectrum licence confers a legal monopoly on the licensee for the duration of its term
- B. A firm may use spectrum without a licence once it registers its equipment directly with the ITU
- C. Spectrum is treated as an abundant resource whose use rarely calls for active regulatory management
- D. A new firm cannot launch a mobile network without spectrum, which is scarce and tightly regulated ✓**

Answer: D. A new firm cannot launch a mobile network without spectrum, which is scarce and tightly regulated

23. Under rate-of-return regulation the regulator sets the allowed profit level, while under price-cap regulation it sets a ceiling on prices. A main advantage of price-cap regulation over rate-of-return is that it:

- A. Gives the firm an incentive to cut costs, since it keeps any efficiency gains as extra profit ✓**
- B. Assures the regulated firm of a return that at least matches its own cost of capital each year
- C. Removes the regulator's need to gather detailed financial data on the firm's underlying costs
- D. Holds retail prices fixed from year to year, thereby offering consumers a degree of certainty

Answer: A. Gives the firm an incentive to cut costs, since it keeps any efficiency gains as extra profit

24. A mobile operator that also owns the towers its competitors must use is said to be vertically integrated. A regulatory concern with vertical integration is that the firm may:

- A. Be unable to price its own retail services competitively unless it receives an ongoing government subsidy to do so
- B. Favour its own downstream retail operations by providing rivals with degraded or overpriced access to the towers ✓**
- C. Reduce the quality of its own retail services, because managing both network and retail stretches it too thinly
- D. Lose much of its market power at the point of integration, as combining the two layers invites closer scrutiny

Answer: B. Favour its own downstream retail operations by providing rivals with degraded or overpriced access to the towers

25. Economic theory defines a public good as one that is non-rival and non-excludable. A broadcast radio signal most closely resembles:

A. A public good ✓

B. A private good, because the broadcaster can charge for each listener

C. A club good, because it is non-rival but perfectly excludable

D. A common resource, because use by one person diminishes supply for others

Answer: A. A public good

26. Structural separation in a telecom market means:

A. The telecom regulator itself divides its own internal functions across several distinct and separately managed government departments

B. Competing operators in the market are required to share their customer databases with one another on demand

C. The monopoly infrastructure (such as the local loop) is owned by a separate company from the retail service providers that use it ✓

D. The government privatises the incumbent operator by selling its shares directly to members of the general public

Answer: C. The monopoly infrastructure (such as the local loop) is owned by a separate company from the retail service providers that use it

27. Accounting separation requires a vertically integrated operator to:

A. Physically and legally split its wholesale and retail operations into two distinct companies, each placed under its own independent management

B. Keep separate financial accounts for its wholesale network and its retail service divisions, as though they were independent entities ✓

C. Share the whole of its detailed internal financial accounts freely with each of its rival operators in the market

D. File one identical set of financial statements across each of the separate countries in which it happens to operate

Answer: B. Keep separate financial accounts for its wholesale network and its retail service divisions, as though they were independent entities

28. A universal service obligation (USO) in telecommunications is best defined as:

A. A requirement that each licensed operator provide all the services it currently offers free of charge to the households it serves

B. An international treaty that bars a signatory country from placing restrictions on its own domestic telecommunications market at will

C. A regulatory requirement that certain basic services are made available to all citizens at an affordable price, regardless of location ✓

D. A licence condition that bars a dominant firm from raising the retail prices of its regulated basic services once these are set

Answer: C. A regulatory requirement that certain basic services are made available to all citizens at an affordable price, regardless of location

29. When the provision of universal service is commercially unviable in high-cost rural areas, regulators commonly establish:

A. A regulated price floor that prevents operators from discounting their services in profitable urban areas

B. A mandatory transfer of spectrum rights from profitable operators to unprofitable rural operators

C. A direct legal obligation requiring consumers themselves to subsidise rural service providers

D. A universal service fund financed by levies on operators in proportion to their revenues ✓

Answer: D. A universal service fund financed by levies on operators in proportion to their revenues

30. Market failure due to information asymmetry occurs when:

A. Both buyers and sellers happen to hold much the same information about the product's underlying quality

B. One side of a market has substantially more information than the other, preventing efficient transactions ✓

- C. The government itself holds more information than the market does about the quantities that should be produced
- D. A single firm sustains lower costs than its rivals over time because of the proprietary technology it owns

Answer: B. One side of a market has substantially more information than the other, preventing efficient transactions

31. A positive externality exists when:

A. A transaction between a buyer and seller generates benefits for third parties who are not part of the transaction ✓

B. A firm knowingly imposes ongoing costs on the wider society that it does not itself have to bear or pay for

C. A government tax deliberately corrects an existing, persistent over-supply of some particular harmful good

D. A monopoly firm simply charges its customers a retail price that sits above the competitive market price

Answer: A. A transaction between a buyer and seller generates benefits for third parties who are not part of the transaction

32. Ex-ante regulation differs from ex-post regulation in that ex-ante:

A. Intervention by the regulator takes place after a breach has already been detected and formally proven to have occurred

B. Regulators choose to lean on general competition law in place of dedicated sector-specific rules and remedies

C. Firms remain free to set their own commercial terms right up until a formal complaint is eventually received

D. Rules are set in advance to shape market behaviour before problems arise, rather than reacting to them after the fact ✓

Answer: D. Rules are set in advance to shape market behaviour before problems arise, rather than reacting to them after the fact

33. An operator designated as having Significant Market Power (SMP) is one that:

A. Simply holds an operating licence that happens to run for a fixed term of fewer than five years in total duration

B. Enjoys a position of economic strength that allows it to behave to an appreciable extent independently of competitors, customers and consumers ✓

C. Reports annual revenues that consistently fall below the national regulatory authority's own official reporting threshold for the sector

D. Operates within remote rural districts where it faces little urban competition and serves a small, dispersed subscriber base

Answer: B. Enjoys a position of economic strength that allows it to behave to an appreciable extent independently of competitors, customers and consumers

34. Before assessing whether an operator has SMP, a regulator must first define the relevant market. The relevant product market is defined as:

A. The complete range of products offered by each licensed operator across the whole of the country

B. Whatever products the regulator personally decides to regulate in pursuit of its own policy aims

C. The smallest set of products that are close enough substitutes to constrain each other's prices ✓

D. Just the services that happen to be covered by the dominant operator's own individual licence

Answer: C. The smallest set of products that are close enough substitutes to constrain each other's prices

35. A 'bottleneck' facility in telecommunications is one that:

A. Sits at the exact geographic centre of the national telecommunications network, midway between its main exchanges

B. May be used by the incumbent operator under the specific terms that are written into the incumbent's own licence

C. Is shared on a voluntary basis by the operators in the market under the terms of a formal co-regulation agreement

D. Is essential for reaching end customers, and which competitors cannot duplicate economically, giving the owner market power ✓

Answer: D. Is essential for reaching end customers, and which competitors cannot duplicate economically, giving the owner market power

36. Local loop unbundling (LLU) allows:

A. Alternative operators to rent the physical copper or fibre pair between the exchange and a customer's premises from the incumbent, at regulated prices ✓

B. The incumbent operator to make use of rival operators' own transmission equipment without paying them a charge for it

C. Individual consumers to take permanent personal ownership of the physical cable that runs between their premises and the exchange

D. Operators to resell one another's retail services to customers without holding a network of their own, buying capacity wholesale instead

Answer: A. Alternative operators to rent the physical copper or fibre pair between the exchange and a customer's premises from the incumbent, at regulated prices

37. Tariff rebalancing in the telecom sector typically refers to:

A. Raising each category of retail price by the same uniform percentage across the board, simply to reflect the annual rate of price inflation

B. Shifting the burden of spectrum licence charges away from the mobile operators so that the fixed-line operators come to bear these costs instead

C. Adjusting prices so that each service covers its cost, reducing cross-subsidies — for instance, raising local call prices and lowering international prices ✓

D. Setting one uniform schedule of retail prices for the operators regardless of each firm's own underlying cost structure or overall size

Answer: C. Adjusting prices so that each service covers its cost, reducing cross-subsidies — for instance, raising local call prices and lowering international prices

38. An operator is likely to retain structural dominance even after market liberalisation largely because it:

A. Has already lost the great majority of its existing subscribers over to the rival firms that recently entered the market

B. Controls legacy infrastructure (such as the local loop or backhaul) that new entrants cannot easily or economically replicate ✓

C. Holds an operating licence that happened to be issued in the period just after the wider market was first opened up

D. Provides very few retail services of its own in the newly opened and now fully competitive downstream retail market

Answer: B. Controls legacy infrastructure (such as the local loop or backhaul) that new entrants cannot easily or economically replicate

39. A regulatory impact assessment (RIA) is designed to:

A. Audit the detailed financial accounts of a regulated company in the aftermath of a breach it has already committed

B. Determine the precise size of the financial penalty to impose on an operator that has been found guilty of market abuse

C. Estimate the costs and benefits of a proposed regulatory intervention before it is implemented, to ensure its net effect is positive ✓

D. Formally replace the entire public consultation process with a single short internal report drafted by the regulator's staff

Answer: C. Estimate the costs and benefits of a proposed regulatory intervention before it is implemented, to ensure its net effect is positive

40. A light-touch regulatory approach is typically characterised by:

A. Minimal intervention, with the regulator stepping in only when there is clear market failure or

a specific complaint ✓

- B. Detailed, highly prescriptive rules that govern each separate aspect of an operator's day-to-day behaviour
- C. The outright nationalisation of the dominant operator, undertaken chiefly to head off future price abuses
- D. Prohibiting rival forms of competition outright in order to protect the position of the existing natural monopoly

Answer: A. Minimal intervention, with the regulator stepping in only when there is clear market failure or a specific complaint

41. Functional separation requires a dominant operator to:

- A. Sell off its entire network infrastructure outright to a genuinely independent and unrelated third-party company at its full market value
- B. Provide the same wholesale prices to retail rivals as it charges its own retail division, well ahead of the point at which formal separation occurs
- C. Establish a distinct universal service fund that is financed out of its own revenues, drawing on no outside contributions or support from others

D. Operate its wholesale access network as a functionally separate business unit with independent management and systems, even if ownership remains unified ✓

Answer: D. Operate its wholesale access network as a functionally separate business unit with independent management and systems, even if ownership remains unified

42. The main argument in favour of infrastructure competition (multiple physical networks) over service competition (multiple retailers on one network) is that it:

A. Leads to greater dynamic efficiency and innovation as firms compete across multiple layers of the value chain ✓

- B. Reduces the total overall cost of building out broadband networks right across the whole of the country
- C. Is readily achievable even in rural areas, where the economics may support just a single viable network
- D. Removes the need for access regulation once the competing physical networks have been fully built out

Answer: A. Leads to greater dynamic efficiency and innovation as firms compete across multiple layers of the value chain

43. Regulators have intervened in international roaming markets because:

A. Roaming agreements freely negotiated between operators are explicitly prohibited under the WTO Basic Telecommunications Agreement

B. Roaming charges were often far above cost, exploiting consumers who had no practical ability to switch provider while abroad ✓

- C. The ITU formally and strictly requires that international calls be routed through a single central global switching hub
- D. Overall consumer demand for roaming services was historically too low to justify commercial provision by operators

Answer: B. Roaming charges were often far above cost, exploiting consumers who had no practical ability to switch provider while abroad

44. When a regulator performs a cost-benefit analysis of a proposed remedy, it should compare:

- A. The regulator's own annual administrative budget against the total yearly revenue earned by the dominant operator in the market
- B. The average income of subscribers in rural districts against the average income of subscribers living in urban districts
- C. The current market share held by each individual operator against the calculated overall national market average

D. The direct costs imposed on industry and consumers against the economic benefits of correcting the market failure ✓

Answer: D. The direct costs imposed on industry and consumers against the economic benefits of

correcting the market failure

45. A multi-sector regulator that oversees telecoms, energy and water simultaneously may have the advantage of:

- A. Focusing its specialist expertise narrowly on the single industry with its own particular technical characteristics
- B. Having a considerably larger administrative staff on hand to investigate each individual complaint that is received
- C. Operating at arm's length from government in a way that separate, sector-specific bodies would find hard to match

D. Applying consistent regulatory principles across sectors and avoiding conflicts between sector-specific agencies ✓

Answer: D. Applying consistent regulatory principles across sectors and avoiding conflicts between sector-specific agencies

46. When two large mobile operators propose to merge, reducing the number of national operators from four to three, the regulator's primary competition concern is likely to be:

- A. An immediate and unavoidable increase in the spectrum licence costs faced by the newly merged entity
- B. A risk that the newly merged entity will prove unable to recover the licence costs it has taken on
- C. A reduction in competitive pressure that could lead to higher prices or lower investment ✓**
- D. A formal requirement that the smaller of the two merged operators buy out the regulator itself

Answer: C. A reduction in competitive pressure that could lead to higher prices or lower investment

47. The main advantage of sector-specific telecom regulation over reliance on general competition law is that:

- A. Sector-specific rules can be calibrated to the industry's technical characteristics and applied proactively, without waiting for a market abuse to occur ✓**
- B. Competition law tends to reach much the same practical outcome as dedicated sector-specific regulation would in a typical case
- C. Sector regulation is inherently more flexible than the alternative, simply because it has no set of prescribed rules for firms to follow
- D. Competition authorities are formally and, in practice, indefinitely forbidden from intervening in the telecom markets under the stated conditions

Answer: A. Sector-specific rules can be calibrated to the industry's technical characteristics and applied proactively, without waiting for a market abuse to occur

48. Asymmetric regulation imposes obligations on the dominant operator that are not applied to smaller rivals. The justification for this asymmetry is that:

- A. Smaller operators are, on the whole, simply managed better and so they need considerably fewer regulatory rules imposed on them
- B. Only the dominant operator has the market power to harm competition or consumers in ways that the market itself cannot correct ✓**
- C. Each operator active in the market ought to be regulated in precisely the same way regardless of the market share it happens to hold
- D. Asymmetric regulation of this specific kind is mandated by the WTO Reference Paper in the great majority of documented cases

Answer: B. Only the dominant operator has the market power to harm competition or consumers in ways that the market itself cannot correct

49. Periodic market reviews, in which the regulator reassesses which operators have SMP and what remedies apply, serve primarily to:

- A. Steadily ratchet up the regulatory obligations that are placed on operators across the whole market as the years go by
- B. Allow the regulator to raise additional annual licence fees specifically from the dominant operators alone
- C. Ensure that regulatory obligations remain proportionate as market conditions evolve, and are removed when no longer needed ✓**

D. Transfer ownership of the incumbent operator back to the state should effective competition fail to develop

Answer: C. Ensure that regulatory obligations remain proportionate as market conditions evolve, and are removed when no longer needed

50. Regulatory forbearance occurs when a regulator:

A. Simply fails to apply its own regulations properly owing to administrative error or chronic under-resourcing

B. Is formally prevented from intervening in the market by a court injunction that the operator itself had obtained

C. Deliberately increases the overall stringency of its regulation in response to changing market conditions

D. Deliberately refrains from exercising its available powers in a market, on the grounds that competition is sufficient ✓

Answer: D. Deliberately refrains from exercising its available powers in a market, on the grounds that competition is sufficient

51. Regulatory independence from the regulated industry encompasses institutional, financial and operational dimensions. Which dimension is concerned with ensuring regulators are not housed within the same organisation as operators?

A. Financial independence

B. Institutional independence ✓

C. Operational independence

D. Political independence

Answer: B. Institutional independence

52. A key risk where the regulator's annual budget is set entirely at the government's discretion is that:

A. The regulator will often over-spend heavily on enforcement against politically connected firms

B. Operators will demand the right to audit the regulator's own accounts directly and regularly

C. The regulator may be starved of resources as an indirect way of constraining its activity ✓

D. The regulator will, in effect, come to depend on the operators it oversees for its financial survival

Answer: C. The regulator may be starved of resources as an indirect way of constraining its activity

53. Which of the following is a mechanism that makes a regulator accountable without compromising its independence?

A. A requirement to publish an annual report explaining its decisions to the legislature and public ✓

B. Allowing the responsible minister to personally overturn a regulatory decision once it has been made

C. Requiring the regulated operator itself to formally sign off on regulatory decisions before they take effect

D. Permitting the regulator to operate freely without publishing the criteria on which its decisions rest

Answer: A. A requirement to publish an annual report explaining its decisions to the legislature and public

54. Transparency in regulation requires that:

A. The entire body of correspondence between the regulator and the operators it regulates is to be kept strictly confidential throughout

B. The regulator shares just its bare final decisions with the public, without publishing the underlying reasoning that led up to them

C. The criteria, process and reasoning behind regulatory decisions are publicly available so that affected parties can understand and challenge them ✓

D. Consultation documents produced by the regulator are circulated to the single dominant operator in the market and to no one else

Answer: C. The criteria, process and reasoning behind regulatory decisions are publicly available so that affected parties can understand and challenge them

55. Regulatory certainty is valued by investors primarily because:

A. Stable rules tend, over time, to translate into steadily lower retail prices for the great majority of consumers

B. Predictable rules reduce the risk premium they require when committing long-term capital to network infrastructure ✓

C. Regulatory certainty of this kind holds back operators that might otherwise enter or exit the market at will

D. Predictable rules do away with much of the need for a formal licensing process to exist in the first place

Answer: B. Predictable rules reduce the risk premium they require when committing long-term capital to network infrastructure

56. Under RPI-X price-cap regulation, if RPI is the retail price index and X is an efficiency factor, operators are required to:

A. Increase their regulated prices by the full RPI each year regardless of their own internal cost savings

B. Hold their regulated prices fixed in nominal terms year after year, regardless of the ongoing rate of inflation

C. Reduce their regulated prices in real terms by X per cent per year, passing efficiency gains to consumers ✓

D. Set their prices at exactly the rate of inflation minus whatever government subsidy happens to apply

Answer: C. Reduce their regulated prices in real terms by X per cent per year, passing efficiency gains to consumers

57. A high X factor in a price-cap formula (RPI-X) implies:

A. The regulator expects the operator to achieve large efficiency savings and requires it to pass them on to consumers quickly ✓

B. The regulator simply expects the wider market to grow steadily less competitive over the course of the coming years

C. The regulated operator will instead receive a direct annual subsidy equal to X per cent of its total yearly revenues

D. The entire price cap will be lifted altogether once a fixed period of exactly X years of regulation has finally passed

Answer: A. The regulator expects the operator to achieve large efficiency savings and requires it to pass them on to consumers quickly

58. A well-known drawback of rate-of-return (RoR) regulation is that it may create an incentive for the regulated firm to:

A. Under-invest in new infrastructure, holding back on capital projects so as to avoid attracting regulatory scrutiny

B. Reduce its retail prices below cost, specifically in order to deliberately drive its rivals out of the market

C. Resist the kinds of technological change that would oblige it to undertake fresh rounds of new capital spending

D. Over-invest in capital (the Averch-Johnson effect), because profit is calculated as a percentage of the asset base ✓

Answer: D. Over-invest in capital (the Averch-Johnson effect), because profit is calculated as a percentage of the asset base

59. A licence condition requiring a mobile operator to achieve a minimum 95 percent population coverage and a maximum 2-second call-setup delay is an example of:

A. Quality-of-service (QoS) regulation imposing minimum performance standards ✓

B. Price-cap regulation limiting how much the operator may charge

C. Accounting separation requiring separate financial records

D. Structural separation requiring distinct ownership of the access network

Answer: A. Quality-of-service (QoS) regulation imposing minimum performance standards

60. A regulator that requires a dominant operator to submit annual regulatory accounts

broken down by network element cost, retail revenue and margin is using:

- A. Accounting separation to improve cost transparency and detect potential cross-subsidies ✓**
- B. Structural separation to legally mandate the creation of two separate and distinct companies
- C. A formal price-cap review carried out specifically to determine the operator's new X factor
- D. A formal market review carried out specifically to define the relevant product market

Answer: A. Accounting separation to improve cost transparency and detect potential cross-subsidies

61. A regulatory sandbox allows:

- A. Operators to remove themselves from most forms of regulation on a voluntary basis for a fixed, set period of time
- B. Governments to informally trial new regulatory agencies in advance of the point at which they are established in law
- C. Firms to test innovative products or business models in a controlled environment with relaxed rules, subject to safeguards ✓**
- D. Regulators to impose considerably higher fines on firms than the underlying legislation would normally permit them to

Answer: C. Firms to test innovative products or business models in a controlled environment with relaxed rules, subject to safeguards

62. A sunset clause in a regulatory instrument provides that:

- A. The regulated firm concerned must fully upgrade its network before a specified statutory deadline arrives
- B. The sitting regulator personally must resign from office once its own five-year term formally ends
- C. The obligations under the licence are instead phased in gradually over the course of several years
- D. The regulation automatically expires after a set time unless the regulator actively decides to renew it ✓**

Answer: D. The regulation automatically expires after a set time unless the regulator actively decides to renew it

63. During a periodic market review, if the regulator finds that effective competition has developed and the dominant operator no longer has SMP, the appropriate response is to:

- A. Remove or relax the sector-specific obligations previously imposed on that operator ✓**
- B. Impose considerably stricter conditions specifically to safeguard the competition that has newly emerged
- C. Transfer the operator's own spectrum holdings directly to a separate, state-owned entity instead
- D. Launch a full formal inquiry to determine whether outright re-nationalisation of the firm is needed

Answer: A. Remove or relax the sector-specific obligations previously imposed on that operator

64. When an interconnection agreement cannot be reached through commercial negotiation, most regulatory frameworks allow either party to:

- A. Withdraw wholesale services from the rival operator until an agreement is finally reached
- B. Refer the dispute to the regulator for binding arbitration or determination ✓**
- C. Seek damages directly from the rival operator's own shareholders in a consumer court
- D. Delay interconnection indefinitely, without facing a penalty for the resulting hold-up

Answer: B. Refer the dispute to the regulator for binding arbitration or determination

65. Regulatory risk, from the perspective of an investor in telecoms infrastructure, refers to:

- A. The physical risk that the operator's network equipment might be seriously damaged by extreme weather events
- B. The risk that the regulator will change the rules in ways that reduce the expected return on invested capital ✓**
- C. The risk that existing subscribers will switch across to rival operators once newer services are launched
- D. The risk that competition law will end up prohibiting the operator from charging its cost-based prices

Answer: B. The risk that the regulator will change the rules in ways that reduce the expected return on invested capital

66. Monitoring and enforcement are distinct regulatory activities. Monitoring is primarily concerned with:

- A. Imposing fines and other penalties on firms that have already formally breached their own licences
- B. Deciding in the first place what specific obligations should be imposed on operators generally
- C. Publishing detailed annual reports covering the regulator's own internal financial performance
- D. Collecting data and tracking compliance with regulatory obligations before a breach has occurred ✓**

Answer: D. Collecting data and tracking compliance with regulatory obligations before a breach has occurred

67. A regulator that responds to a minor licence breach with an informal warning, escalating to a formal compliance notice and then a fine if the breach persists, is applying:

- A. A graduated (or proportionate) enforcement approach ✓**
- B. Regulatory capture, whereby the regulator avoids penalising the industry
- C. Co-regulation, where the industry jointly decides on the appropriate penalty
- D. Zero regulation, since no immediate fine is imposed on the first breach

Answer: A. A graduated (or proportionate) enforcement approach

68. A voluntary code of practice adopted by operators without legislative force serves primarily to:

- A. Replace the formal licence conditions in their entirety with a set of non-binding voluntary commitments made in their place
- B. Establish baseline standards of consumer treatment that operators commit to publicly, creating reputational incentives for compliance ✓**
- C. Allow the regulator to impose fines for a breach without first having to work through the usual steps of due process
- D. Transfer the whole of the regulatory responsibility for the matter across to the ordinary courts of law in its place

Answer: B. Establish baseline standards of consumer treatment that operators commit to publicly, creating reputational incentives for compliance

69. When a regulator proposes a new interconnection pricing methodology, best practice requires it to publish a consultation document and allow a period for:

- A. The responsible government minister to formally approve the document before it is sent out to the operators
- B. The single dominant operator alone to veto proposals that it personally finds commercially disadvantageous
- C. Operators, consumers and other interested parties to comment before the methodology is finalised ✓**
- D. The courts to formally pre-approve the methodology as being legally valid before its scheduled publication date

Answer: C. Operators, consumers and other interested parties to comment before the methodology is finalised

70. In developing economies, one frequently cited obstacle to effective regulation is:

- A. An overall excess of regulatory staff, whose collective activity imposes unnecessary administrative burdens on each licensed operator in the sector
- B. The knowledge and information gap between the regulator and the companies it regulates, due to limited technical and financial expertise in the regulatory body ✓**
- C. The outright refusal of the incumbent operators to obey the governing law, whatever the particular circumstances they happen to face
- D. The complete and total absence of legislation of the kind needed to enable the creation of a regulatory body in the very first place

Answer: B. The knowledge and information gap between the regulator and the companies it regulates, due to limited technical and financial expertise in the regulatory body

71. Why are licence conditions a particularly powerful regulatory instrument compared with general guidelines?

- A. Because informal guidelines tend to be published in the wake of a court order, whereas formal licences issue much faster
- B. Because binding licence conditions apply, without exception, to each firm operating across the sectors of the economy
- C. Because non-binding guidelines must first be approved by the ITU, whereas formal licences need no such prior approval
- D. Because they are legally binding on the licence holder and breach can result in enforcement action or licence revocation ✓**

Answer: D. Because they are legally binding on the licence holder and breach can result in enforcement action or licence revocation

72. A regulatory requirement for operators to maintain a formal written complaint procedure and respond within a set number of days is designed to:

- A. Prevent ordinary consumers from taking their complaints directly to a court or an independent tribunal
- B. Ensure consumers have an accessible redress mechanism and operators take service quality seriously ✓**
- C. Transfer consumer protection responsibilities away from the regulator and over to the operator in its place
- D. Allow the operator to legitimately charge consumers a fee for the internal processing of their complaints

Answer: B. Ensure consumers have an accessible redress mechanism and operators take service quality seriously

73. Regulators in smaller markets often benchmark their rules and tariff levels against those in comparable countries. The primary purpose of such benchmarking is to:

- A. Simply copy the exact rules used by the single largest economy, taking little account of local market conditions back at home
- B. Do away with the whole of the national public consultation process and put an international one in its place altogether
- C. Draw on evidence from multiple markets to set well-informed, defensible regulatory parameters in the absence of sufficient domestic data ✓**
- D. Allow individual operators themselves to freely choose whichever country's regulatory framework they would personally prefer

Answer: C. Draw on evidence from multiple markets to set well-informed, defensible regulatory parameters in the absence of sufficient domestic data

74. In most countries, the telecommunications regulatory framework assigns the legislature (parliament) the role of:

- A. Setting the wholesale interconnection prices individually, on a strict case-by-case basis, each time they arise
- B. Enacting the primary legislation that establishes the regulatory body and confers its powers ✓**
- C. Issuing individual operator licences directly and personally handling the spectrum allocation decisions itself
- D. Investigating each individual consumer complaint that is raised about the operators' billing practices

Answer: B. Enacting the primary legislation that establishes the regulatory body and confers its powers

75. An independent regulatory authority is often governed by a multi-member board rather than a single regulator. The main benefit of a board structure is that:

- A. A multi-member board can issue licences considerably faster because each member simply signs separately
- B. Individual board members are personally and legally liable for regulatory decisions, which deters bias
- C. A multi-member board structure needs no underlying legislation of its own in order to be established
- D. Decisions represent a range of views and are less vulnerable to the bias or capture of any single individual ✓**

Answer: D. Decisions represent a range of views and are less vulnerable to the bias or capture of any single individual

76. Spectrum refarming refers to the process of:

A. Reassigning spectrum bands previously used by one technology (e.g. 2G) to a newer, more spectrally efficient technology (e.g. 4G or 5G) ✓

B. Allocating brand-new spectrum bands that were identified for the first time through recent satellite observation work

C. Transferring spectrum rights on a lasting basis between two or more separate countries under a formal, signed ITU agreement

D. Increasing the total quantity of MHz available within a given band, simply by cutting down the size of its guard bands further

Answer: A. Reassigning spectrum bands previously used by one technology (e.g. 2G) to a newer, more spectrally efficient technology (e.g. 4G or 5G)

77. In spectrum secondary trading, spectrum rights are:

A. Handed straight back to the government for cost-free reallocation once the licensee no longer has a use for them

B. Auctioned directly by the ITU itself to the single highest bidder drawn from across its various member states

C. Transferred or leased from one licence holder to another through market transactions, subject to regulator approval ✓

D. Assigned by default to whichever operator happens to hold the longest network operating history on record

Answer: C. Transferred or leased from one licence holder to another through market transactions, subject to regulator approval

78. Cognitive radio technology enables more efficient spectrum use primarily by:

A. Physically expanding the total radio spectrum available worldwide by opening up brand-new, previously untapped frequency ranges for general use

B. Allowing devices to sense and opportunistically use spectrum bands that are temporarily idle, without causing harmful interference to primary users ✓

C. Requiring the operators in the market to share a single licensed frequency band between themselves on a lasting, ongoing basis

D. Replacing the existing licensed spectrum bands with unlicensed, licence-exempt bands right across the range of telecom services

Answer: B. Allowing devices to sense and opportunistically use spectrum bands that are temporarily idle, without causing harmful interference to primary users

79. A regulator may impose spectrum caps (or spectrum limits) on individual operators in order to:

A. Ensure that each operator active in the market ends up holding roughly the same overall amount of licensed spectrum

B. Reduce the total quantity of spectrum that is put to use by the operators taken together across the country

C. Let the incumbent operator retain the bulk of the remaining spectrum once market liberalisation has occurred

D. Prevent any single operator from acquiring so much spectrum that it can disadvantage rivals, particularly in auctions ✓

Answer: D. Prevent any single operator from acquiring so much spectrum that it can disadvantage rivals, particularly in auctions

80. When a country's radio transmissions could cause interference to services in a neighbouring country, the coordination process is managed through:

A. Bilateral and multilateral agreements, guided by the ITU Radio Regulations ✓

B. The World Trade Organization's Basic Telecommunications Agreement, negotiated between member states

C. A dedicated telecommunications sub-committee that operates under the UN Security Council itself

D. The National Regulatory Authority of the affected country acting on its own initiative, in a unilateral fashion

Answer: A. Bilateral and multilateral agreements, guided by the ITU Radio Regulations

81. In the E.164 international numbering plan, a full international telephone number comprises:

A. A network code followed directly by a subscriber number, with an overall minimum of ten digits

B. A two-letter country prefix followed by both an area code and a separate subscriber number

C. A country code followed by the national (significant) number, with the total not exceeding 15 digits ✓

D. A six-digit operator code followed directly by the individual subscriber's own national number

Answer: C. A country code followed by the national (significant) number, with the total not exceeding 15 digits

82. Non-geographic numbers differ from geographic numbers in that they:

A. Are not tied to a specific physical location, and are used for services such as freephone, premium-rate, or shared-cost calls ✓

B. May be assigned to mobile subscribers specifically, and are withheld from customers of the fixed-line operators

C. Must begin with the full international country code of whichever network happens to originate the particular call in question

D. Are set aside for the sole use of the emergency services, following a common arrangement adopted across countries

Answer: A. Are not tied to a specific physical location, and are used for services such as freephone, premium-rate, or shared-cost calls

83. A freephone (toll-free) service number is one where:

A. Neither the caller nor the called party pays a penny, since the government fully subsidises the cost of the call

B. The caller instead pays a considerably higher-than-standard rate, which the operator itself then retains

C. The number in question is available just to emergency services and hospitals, and to nobody besides them

D. The cost of the call is borne by the called party (the business) rather than the calling consumer ✓

Answer: D. The cost of the call is borne by the called party (the business) rather than the calling consumer

84. Premium-rate service numbers expose consumers to the risk of high unintended charges. A common regulatory response is to:

A. Prohibit the use of premium-rate numbers outright, right across the whole of the retail market

B. Require clear price disclosure before the consumer connects and impose a maximum per-minute or per-call cap ✓

C. Require that premium-rate calls be paid for in full and in advance by the calling consumer

D. Let operators set their own premium rates freely, with no obligation to disclose them to callers

Answer: B. Require clear price disclosure before the consumer connects and impose a maximum per-minute or per-call cap

85. Short codes (e.g. 3- to 6-digit numbers) are administered by national regulators because:

A. Short codes are instead allocated by the ITU directly to each individual country as one part of the wider E.164 numbering plan

B. They are a scarce resource — their brevity limits how many exist — and must be allocated fairly across competing service providers ✓

C. Short codes are effectively abundant in overall supply, and for that reason they call for little in the way of formal administration

D. Short codes are reserved for use by government agencies and by the telecom regulators themselves, rather than by service providers

Answer: B. They are a scarce resource — their brevity limits how many exist — and must be

allocated fairly across competing service providers

86. In a standard telephone call, the originating operator is the one that:

- A. Receives the call on behalf of the called subscriber and is the network that actually rings their handset
- B. Issues the specific telephone number at which the called party can ultimately be reached by other callers
- C. Carries the call from the calling subscriber and delivers it to the network of the terminating operator ✓**
- D. Determines the final retail price that is actually charged to the called party for receiving the call

Answer: C. Carries the call from the calling subscriber and delivers it to the network of the terminating operator

87. Mobile termination rates (MTRs) are regulated in many countries because:

- A. Mobile operators, as a general rule, charge their own subscribers far too little money for the incoming calls that they receive
- B. The ITU formally and explicitly mandates that mobile termination rates be set at a level of zero across its member states
- C. Mobile termination rates of this kind are paid directly by the government to operators, rather than between the operators
- D. The terminating operator has a monopoly over access to its own subscribers, allowing it to charge above-cost rates to originating operators ✓**

Answer: D. The terminating operator has a monopoly over access to its own subscribers, allowing it to charge above-cost rates to originating operators

88. Long-run incremental cost (LRIC) is used as a basis for regulated interconnection prices because:

- A. It simply measures the actual historic accounting cost that the operator had already spent at some earlier point in the past
- B. It secures for the regulated operator a fixed, government-mandated profit margin, applied specifically to its interconnection traffic
- C. It is simply the particular costing model that the WTO officially prefers for use across cross-border interconnection arrangements
- D. It reflects the efficient forward-looking cost of providing the specific service, excluding costs unrelated to the traffic being terminated ✓**

Answer: D. It reflects the efficient forward-looking cost of providing the specific service, excluding costs unrelated to the traffic being terminated

89. Retail-minus pricing sets the wholesale access price at:

- A. A wholesale access price set quite deliberately well above the incumbent's own retail price, so as to deter the entry of competitors
- B. The incumbent's retail price minus the retail costs (marketing, billing, etc.) it avoids when selling wholesale, ensuring viable retail competition ✓**
- C. Zero, on the basis that wholesale network access to the incumbent's network should be provided free of charge to the operators that seek it
- D. The full and complete cost of building a brand-new, fully competing rival network right from scratch, recalculated on each occasion

Answer: B. The incumbent's retail price minus the retail costs (marketing, billing, etc.) it avoids when selling wholesale, ensuring viable retail competition

90. Domestic roaming agreements allow a mobile operator to:

- A. Offer its own subscribers calls at no charge whenever they make an international visit to a designated partner country
- B. Make lasting use of a foreign operator's own spectrum inside its national borders, with no fixed limit set on the term
- C. Charge its own subscribers a noticeably lower price whenever they happen to be located within their home region
- D. Provide services to its subscribers in geographic areas it does not cover, by using the**

network of another domestic operator ✓

Answer: D. Provide services to its subscribers in geographic areas it does not cover, by using the network of another domestic operator

91. Regulators impose wholesale roaming caps on visited-network operators. The effect of a wholesale roaming cap is to:

A. Prevent home-network operators from being in a position to sign roaming agreements with other operators, whatever the circumstances

B. Allow subscribers to make free use of whichever network they personally wish, without needing permission from the operators involved

C. Limit the price that visited-network operators may charge the home-network operator, which can then pass lower retail prices to travelling subscribers ✓

D. Set the exact retail price that is charged to consumers by the home operator directly, in a strictly unilateral fashion, without exception

Answer: C. Limit the price that visited-network operators may charge the home-network operator, which can then pass lower retail prices to travelling subscribers

92. A Mobile Virtual Network Operator (MVNO) differs from a full Mobile Network Operator (MNO) in that an MVNO:

A. Owns its own radio spectrum and base stations outright, but chooses not to maintain a retail customer base of its own

B. Operates within remote rural districts, in the specific areas where the full MNOs happen to have no network coverage of their own

C. Is licensed under a general class licence that is free of the various conditions attached to the operating licence of a full MNO

D. Does not own radio spectrum or a radio access network, and instead buys wholesale network access from an MNO to sell services to end customers ✓

Answer: D. Does not own radio spectrum or a radio access network, and instead buys wholesale network access from an MNO to sell services to end customers

93. Passive infrastructure sharing involves operators sharing:

A. Physical structures such as towers, masts, ducts and shelters, without sharing the active radio or transmission equipment ✓

B. The exact same radio frequencies and active antennas installed together on one single, shared cell site

C. Spectrum licences themselves, so that both operators end up transmitting together on one single common frequency band

D. Customer billing systems and subscriber management platforms used internally and separately by each individual operator

Answer: A. Physical structures such as towers, masts, ducts and shelters, without sharing the active radio or transmission equipment

94. Radio Access Network (RAN) sharing goes beyond passive sharing in that operators share:

A. The active radio equipment (antennas, transceivers, controllers) in addition to towers and other physical structures ✓

B. Just the masts and the shelters themselves, but not the associated electronic radio equipment mounted on top of them

C. Core network elements, such as the central switching and routing infrastructure that is used by each of the operators

D. Spectrum licences, so that each operator taking part in the shared RAN transmits on one common frequency band

Answer: A. The active radio equipment (antennas, transceivers, controllers) in addition to towers and other physical structures

95. Net neutrality is the principle that:

A. Officially designated neutral third parties are the sole entities permitted to operate the internet exchange points

B. Internet service providers must provide free-of-charge access to the official government websites and to those alone

C. Internet service providers must treat all internet traffic equally, without discriminating based on source, destination or content type ✓

D. Content companies must instead pay individual consumers directly for the privilege of consuming their own online services

Answer: C. Internet service providers must treat all internet traffic equally, without discriminating based on source, destination or content type

96. In mobile number portability (MNP), the 'losing' operator is:

A. The operator that is actively gaining the subscriber who wishes to keep hold of their existing number

B. The operator from which the subscriber is porting away — the one that currently holds the number ✓

C. The operator responsible for routing the ported call once the entire porting process is complete

D. The operator that centrally administers the national number portability database for the whole industry

Answer: B. The operator from which the subscriber is porting away — the one that currently holds the number

97. Regulators that set maximum porting timelines (e.g. one working day) do so primarily to:

A. Prevent operators from using slow porting processes to discourage subscribers from switching ✓

B. Allow operators to complete the technical work involved before the customer becomes aware of the port

C. Ensure that porting requests are, as a matter of routine, processed strictly in the order received

D. Limit the number of porting requests that an operator has to handle over the course of a given month

Answer: A. Prevent operators from using slow porting processes to discourage subscribers from switching

98. A short licence duration (e.g. five years) compared with a long one (e.g. twenty years) gives the regulator more:

A. Greater certainty for investors who specifically need predictable returns over a much longer period

B. A longer interval before a formal regulatory review of the operator's own conduct falls due for hearing

C. Opportunities to revise the licence conditions as market and technology conditions change ✓

D. Additional spectrum allocation rights that can then be distributed out to brand-new market entrants

Answer: C. Opportunities to revise the licence conditions as market and technology conditions change

99. When renewing a licence, a regulator may attach new or revised conditions. The main constraint on this power is that:

A. No fresh conditions of the kind in question may be added to an existing licence once it has first been granted to the operator

B. The ITU itself is the sole body permitted to formally approve fresh conditions that come to be added at the point of a licence renewal

C. New conditions must be proportionate, justified by current market circumstances, and the operator must have fair notice and an opportunity to respond ✓

D. The national government treasury must separately approve each individual new condition, assessing it for its own fiscal impact first

Answer: C. New conditions must be proportionate, justified by current market circumstances, and the operator must have fair notice and an opportunity to respond

100. A regulatory requirement that all licensed operators connect emergency calls free of charge, even when a subscriber has no credit, is justified on the basis that:

A. Emergency calls actually generate significant revenue that fully offsets their own zero-price status

B. Emergency call access is a fundamental public interest obligation that overrides commercial

pricing considerations ✓

C. The WTO's Basic Telecommunications Agreement explicitly prohibits the levying of a charge for emergency service calls

D. Subscribers who personally call emergency services are later refunded directly by the government itself

Answer: B. Emergency call access is a fundamental public interest obligation that overrides commercial pricing considerations

Part 3

1. A streaming operator raises the monthly price of its video subscription and finds that the number of active subscriptions falls, with incomes and tastes unchanged. This outcome most directly illustrates:

A. The law of supply, where a higher price raises the quantity firms are willing to supply

B. The law of demand, where a higher price reduces the quantity demanded, other things equal ✓

C. A leftward shift of the demand curve brought about by the change in the good's own price

D. A rise in demand elasticity that pushes the operator's total revenue upward as price rises

Answer: B. The law of demand, where a higher price reduces the quantity demanded, other things equal

2. Average household incomes in a region rise and, at every price, subscribers now buy more fixed broadband than before. For broadband treated as a normal good, this is best shown on a diagram as:

A. A rightward shift of the entire demand curve ✓

B. A movement upward along the existing demand curve

C. A movement downward along the existing demand curve

D. A leftward shift of the entire supply curve

Answer: A. A rightward shift of the entire demand curve

3. Subscribers tend to buy a smartphone handset and a mobile data bundle together, so a fall in handset prices raises the quantity of data bundles bought. In demand analysis the two products are best classified as:

A. Substitute goods

B. Complementary goods ✓

C. Inferior goods

D. Independent goods

Answer: B. Complementary goods

4. The market supply curve for a good is drawn holding other influences constant. Among the determinants of the quantity supplied, the variable placed at the centre of the basic supply relationship is the:

A. Price of the good itself ✓

B. Level of consumer income in the market

C. Prices of goods that consumers regard as substitutes

D. Tastes and preferences of the buyers in the market

Answer: A. Price of the good itself

5. A component manufacturer offers more units for sale as the market price rises and fewer as it falls, with technology and input costs unchanged. This positive price-quantity relationship is captured by the:

A. Law of demand

B. Principle of diminishing marginal returns

C. Law of supply ✓

D. Concept of consumer surplus

Answer: C. Law of supply

6. Engineers introduce a cheaper production technique that lowers the cost of making network routers. Holding the market price constant, standard analysis predicts that the supply curve for routers will:

- A. Shift to the right, as producers offer more at each price ✓
- B. Shift to the left, as producers offer less at each price
- C. Rotate until it becomes perfectly vertical at the old quantity
- D. Stay fixed while buyers move downward along the demand curve

Answer: A. Shift to the right, as producers offer more at each price

7. Regulators observe that when the price of a basic voice service rises by 10 percent the quantity demanded falls by just 2 percent. The demand for this service is best described as:

- A. Elastic, because quantity demanded responds sharply to the price change
- B. Unit elastic, because quantity demanded falls in step with the price change
- C. Inelastic, because quantity demanded responds little to the price change ✓
- D. Perfectly elastic, because quantity demanded collapses to zero after the change

Answer: C. Inelastic, because quantity demanded responds little to the price change

8. For a service with many close substitutes, a small price increase leads consumers to cut back their purchases sharply. The demand for such a service is best described as:

- A. Inelastic
- B. Perfectly inelastic
- C. Unit elastic
- D. Elastic ✓

Answer: D. Elastic

9. At the current market price of a good, the quantity that suppliers offer exceeds the quantity that buyers wish to purchase. Standard analysis predicts that the price will tend to:

- A. Rise further above the equilibrium, widening the excess supply
- B. Fall toward the equilibrium, clearing the excess supply ✓
- C. Stay fixed, since a surplus has no effect on the market price
- D. Fall below the equilibrium and settle at a lasting shortage

Answer: B. Fall toward the equilibrium, clearing the excess supply

10. To keep a service affordable, a regulator caps its retail price below the market equilibrium and holds it there. Standard analysis predicts that this binding price ceiling will most likely produce:

- A. A surplus, because quantity supplied exceeds quantity demanded at the capped price
- B. A fresh equilibrium that settles exactly at the level of the capped price
- C. A rise in supply large enough to remove the gap between demand and supply
- D. A shortage, because quantity demanded exceeds quantity supplied at the capped price ✓

Answer: D. A shortage, because quantity demanded exceeds quantity supplied at the capped price

11. A government guarantees producers a minimum price set above the market equilibrium and commits to hold it there. The most likely consequence of this binding price floor is:

- A. A shortage, because quantity demanded exceeds quantity supplied at the floor price
- B. A swift movement of the market back to its original equilibrium price
- C. A surplus, because quantity supplied exceeds quantity demanded at the floor price ✓
- D. A fall in the quantity supplied to a level beneath the quantity demanded

Answer: C. A surplus, because quantity supplied exceeds quantity demanded at the floor price

12. In public finance, the phrase tax incidence is used specifically to describe:

- A. How the economic burden of a tax ends up divided between buyers and sellers ✓
- B. The statutory rate that lawmakers write into the text of the tax legislation
- C. The total revenue a government expects to raise from imposing the tax
- D. The paperwork and collection costs a tax authority bears in administering the tax

Answer: A. How the economic burden of a tax ends up divided between buyers and sellers

13. A per-unit tax is levied on the buyers of a good rather than on its sellers. In the standard supply-and-demand model, the initial effect is represented as:

- A. A rightward shift of the demand curve
- B. A leftward shift of the supply curve
- C. A rotation that makes the supply curve more elastic
- D. A leftward (downward) shift of the demand curve ✓**

Answer: D. A leftward (downward) shift of the demand curve

14. An identical per-unit tax is placed on two goods. Good X has highly elastic demand while good Y has highly inelastic demand. Compared with buyers, sellers will tend to bear the larger share of the tax on:

- A. Good X, where demand is elastic ✓**
- B. Good Y, where demand is inelastic
- C. Both goods in equal proportion, whatever the elasticities
- D. Neither good, since the duty to remit the tax rests with buyers

Answer: A. Good X, where demand is elastic

15. An economy in which a central authority owns the major resources and directs what is produced, how, and for whom, is best described as a:

- A. Traditional economy, coordinated by inherited custom
- B. Market economy, coordinated mainly through prices
- C. Mixed economy, blending markets with state action
- D. Command economy, coordinated by central planning ✓**

Answer: D. Command economy, coordinated by central planning

16. An economy in which production methods and the allocation of goods follow inherited customs, with community elders having the final say, is best described as a:

- A. Command economy
- B. Market economy
- C. Mixed economy
- D. Traditional economy ✓**

Answer: D. Traditional economy

17. Supporters of the market economy argue that when buyers and sellers each pursue their own interest, competition and prices tend to guide resources toward valued uses. This coordinating tendency is often referred to as the:

- A. Invisible hand of the market ✓**
- B. Central plan of the state
- C. Fixed rule of inherited custom
- D. Visible hand of the regulator

Answer: A. Invisible hand of the market

18. A defining assumption of a perfectly competitive market is that:

- A. A few large firms sell differentiated products and shade prices to match rivals
- B. Many firms sell an essentially identical product, so no single firm sets the price ✓**
- C. A single firm supplies the whole market and searches for its profit-maximising price
- D. The state sets a common price that all the firms in the market then charge

Answer: B. Many firms sell an essentially identical product, so no single firm sets the price

19. Because a monopolist faces the whole market demand curve rather than a going market price, it is best described as a:

- A. Price taker, which accepts the price set by the wider market
- B. Quantity taker, which accepts an output level fixed by its rivals
- C. Price maker, which chooses price and quantity along the demand curve ✓**
- D. Cost taker, which accepts input prices dictated by the regulator

Answer: C. Price maker, which chooses price and quantity along the demand curve

20. An industry is described as a natural monopoly when, over the relevant range of output:

- A. Several competing firms can serve the market more cheaply than one, owing to rivalry
- B. The state licenses a single firm and bars new entrants from the market by statute
- C. Firms coordinate their output so that the market behaves as though one firm supplied it, even while rivals remain active
- D. One firm can serve the whole market at lower cost than several firms could, owing to economies of scale ✓**

Answer: D. One firm can serve the whole market at lower cost than several firms could, owing to economies of scale

21. Where a genuine natural monopoly supplies an essential service, a common regulatory response is to allow the single firm to remain but to:

- A. Require several firms to duplicate the network regardless of the added cost
- B. Step back from oversight and let the firm price the service as it sees fit
- C. Regulate its prices and conduct so outcomes move closer to the competitive result ✓**
- D. Take the firm and its rivals across the economy into full public ownership

Answer: C. Regulate its prices and conduct so outcomes move closer to the competitive result

22. A factory sells its output cheaply but discharges untreated waste that harms downstream communities whose losses it does not pay for. This gap between private and social cost is an example of a market failure known as a:

- A. Positive externality
- B. Public good
- C. Price ceiling
- D. Negative externality ✓**

Answer: D. Negative externality

23. On a supply-and-demand diagram, the area between the demand curve and the market price, up to the quantity traded, measures:

- A. Consumer surplus, the gain buyers get above what they actually pay ✓**
- B. Producer surplus, the gain sellers get above their marginal cost
- C. Tax revenue, the amount the government raises from the market
- D. Deadweight loss, the value of trades that fail to take place

Answer: A. Consumer surplus, the gain buyers get above what they actually pay

24. Compared with the competitive outcome, a profit-maximising monopoly restricts output and raises price. The value of the mutually beneficial trades that are lost as a result is referred to as the:

- A. Producer surplus captured by the monopoly
- B. Deadweight loss caused by the monopoly ✓**
- C. Consumer surplus retained by the buyers
- D. Tax incidence shared across the market

Answer: B. Deadweight loss caused by the monopoly

25. Because society's resources are limited while people's wants are not, every choice to use resources one way forgoes the next-best alternative. Economists call the value of that forgone alternative the:

- A. Opportunity cost of the choice ✓**
- B. Marginal revenue of the choice
- C. Sunk cost of the choice
- D. Fixed cost of the choice

Answer: A. Opportunity cost of the choice

26. In many countries before the 1980s, telephone services were run by a government post and telecommunications administration (a PTT). This arrangement meant the service was typically provided by:

- A. A group of competing private carriers

B. A set of independent regional cooperatives

C. A state-owned monopoly ✓

D. A foreign-owned equipment consortium

Answer: C. A state-owned monopoly

27. A frequently cited economic reason for the early telecom monopoly was that the high fixed cost of building a network, set against limited demand, made it hard for a second entrant to recover its outlay. This gave the industry the character of a:

A. Perfectly contestable market

B. Natural monopoly ✓

C. Fully deregulated open market

D. Traditional subsistence economy

Answer: B. Natural monopoly

28. Beyond economics, a political reason governments kept telecommunications in state hands was that they:

A. Were barred by an international postal treaty from licensing a private telephone operator

B. Had no ministry with the technical means to oversee a basic national postal service

C. Regarded communications as strategically sensitive and wished to keep them under state control ✓

D. Preferred to sell radio-spectrum rights straight to the highest private bidder of the day

Answer: C. Regarded communications as strategically sensitive and wished to keep them under state control

29. In the telecom sector, the term liberalization is best understood as:

A. The sale of a state operator's ownership to private shareholders

B. The setting of shared technical standards across the operators

C. The intervention of the state to reach particular market outcomes

D. The move from a monopoly market toward one open to competition ✓

Answer: D. The move from a monopoly market toward one open to competition

30. When a regulator removes the incumbent's exclusive rights to provide certain services, the most direct effect of that step is to:

A. Move ownership of the incumbent's shares into private investors' hands

B. Let new entrants offer services that had been reserved for the incumbent ✓

C. Grant the incumbent a longer, protected and exclusive market position

D. Fix a strict cap on the retail price of each of the incumbent's services

Answer: B. Let new entrants offer services that had been reserved for the incumbent

31. In the telecom sector, the term privatization refers specifically to:

A. Moving the ownership of an operator from the state to private hands ✓

B. Opening a monopoly market to competition from new entrants

C. Placing universal-service duties on the licensed operators

D. Establishing a new and independent national regulatory body to supervise the sector

Answer: A. Moving the ownership of an operator from the state to private hands

32. A broad public offering of shares to many investors, rather than a sale to a single strategic buyer, has been the more common route to privatizing an incumbent in:

A. Developing countries with thin capital markets

B. Countries that never had an incumbent operator

C. Countries retaining a full state telephone monopoly

D. Highly industrialised countries ✓

Answer: D. Highly industrialised countries

33. A country sells a majority stake in its incumbent fixed-line operator to private investors but continues to grant that operator exclusive rights over fixed services. This is best described as:

- A. Liberalization of the market without privatization of the incumbent
- B. Privatization of the incumbent without liberalization of the market ✓**
- C. Both full liberalization and full privatization together
- D. Neither privatization of the incumbent nor liberalization of the market

Answer: B. Privatization of the incumbent without liberalization of the market

34. In the context of the telecom sector, regulation is best defined as:

- A. The full withdrawal of the government from the market's activity
- B. The transfer of state-owned network assets into private ownership
- C. Government intervention in a market to bring about particular desired outcomes ✓**
- D. The routine self-policing of firms that operate without external oversight of their conduct

Answer: C. Government intervention in a market to bring about particular desired outcomes

35. Regulation is often said to take three basic forms. Setting common interconnection and equipment standards that let different operators' networks work together is an example of:

- A. Public policy regulation
- B. Competition regulation
- C. Voluntary self-regulation
- D. Technical standardization ✓**

Answer: D. Technical standardization

36. Requiring operators to carry emergency calls and to meet content rules judged to serve the public interest is an example of which basic form of regulation?

- A. Public policy regulation ✓**
- B. Technical standardization
- C. Competition regulation
- D. Co-regulation

Answer: A. Public policy regulation

37. The basic form of regulation aimed at establishing and preserving fair rivalry between firms, and fair dealing between firms and their customers, is:

- A. Technical standardization
- B. Public policy regulation
- C. Competition regulation ✓**
- D. Zero regulation

Answer: C. Competition regulation

38. Although telecom regulation can take several forms, the central focus of contemporary regulation of the telecommunications services industry is:

- A. The harmonising of accounting standards across unrelated industries
- B. The creation and maintenance of effective, competitive markets ✓**
- C. The control of socially sensitive products such as alcohol
- D. The removal of private ownership from the telecom sector

Answer: B. The creation and maintenance of effective, competitive markets

39. A widely recognised objective of telecom regulation is to protect the interests of consumers, for example by:

- A. Securing a fixed profit margin for each of the licensed operators
- B. Steering connectivity toward the most commercially attractive urban districts of the country
- C. Addressing poor service quality and unfair terms that individual users struggle to resist ✓**
- D. Holding back new operators that might enter and disturb the market

Answer: C. Addressing poor service quality and unfair terms that individual users struggle to resist

40. Where competitive markets are absent or have failed, a recognised objective is for the regulator to:

- A. Guard against abuses of market power, such as excessive pricing by a dominant firm ✓**

- B. Shelter the dominant firm from the entry of potential competitors
- C. Pull back from oversight and let the market repair itself unaided
- D. Set every operator's retail prices at one identical level right across the national market

Answer: A. Guard against abuses of market power, such as excessive pricing by a dominant firm

41. Which statement best captures the distinction between policy and law drawn in the course material?

- A. Policy can compel or forbid conduct, whereas law merely offers broad guidance that binds no one
- B. Policy and law mean the same thing, so the two terms can be swapped freely
- C. Policy sets the long-term vision while law simply enforces it through the courts

D. Law can compel or forbid conduct, whereas policy sets a direction that guides action toward a goal ✓

Answer: D. Law can compel or forbid conduct, whereas policy sets a direction that guides action toward a goal

42. Within the policy-to-practice chain, the role of regulation is chiefly to:

- A. Turn the objectives set by policy into practical rules, decisions and day-to-day oversight ✓**
- B. Define the broad, long-term national vision that takes shape well before policy itself is formed
- C. Take the place of the underlying legislation and statutory authority
- D. Review the personal finances of government ministers each year

Answer: A. Turn the objectives set by policy into practical rules, decisions and day-to-day oversight

43. A frequently noted problem in the pre-reform arrangement was that one government body often acted at the same time as policy maker, regulator and operator of the telephone service. The main concern with this was that it:

- A. Left the day-to-day running of the telephone service without effective government supervision
- B. Created conflicts of interest that made even-handed treatment of rivals difficult ✓**
- C. Prevented the state from raising revenue through licence fees
- D. Handed effective control of the sector to foreign carriers

Answer: B. Created conflicts of interest that made even-handed treatment of rivals difficult

44. The International Telecommunication Union (ITU) is best described as:

- A. A private club funded by the world's largest mobile network operators
- B. A regional trade bloc that negotiates import tariffs on telecom equipment on behalf of its members
- C. A commercial standards firm owned and directed by major equipment vendors
- D. A United Nations specialised agency dealing with information and communication technologies ✓**

Answer: D. A United Nations specialised agency dealing with information and communication technologies

45. Coordinating the international radio-frequency spectrum and satellite-orbit resources is the responsibility of which ITU sector?

- A. ITU-T (Standardization)
- B. ITU-D (Development)
- C. ITU-P (Policy)
- D. ITU-R (Radiocommunication) ✓**

Answer: D. ITU-R (Radiocommunication)

46. Developing the technical standards that let telecommunications networks and services interwork globally, apart from radio matters, is the responsibility of:

- A. ITU-R (Radiocommunication)
- B. ITU-D (Development)
- C. ITU-T (Standardization) ✓**
- D. ITU-S (Spectrum)

Answer: C. ITU-T (Standardization)

47. Helping to widen equitable, sustainable and affordable access to ICTs, especially in

developing countries, is the principal role of:

- A. ITU-D (Development) ✓**
- B. ITU-R (Radiocommunication)
- C. ITU-T (Standardization)
- D. ITU-A (Access)

Answer: A. ITU-D (Development)

48. The World Trade Organization's Agreement on Basic Telecommunications (the BTA) is chiefly concerned with:

- A. Assigning radio spectrum directly to individual operators in each member state
- B. Setting out rules for competition and encouraging open investment in telecoms across countries ✓**
- C. Fixing the maximum retail tariffs that operators may charge their subscribers
- D. Granting individual operating licences directly to the carriers within each member state's market

Answer: B. Setting out rules for competition and encouraging open investment in telecoms across countries

49. One of the six basic principles of the WTO Reference Paper on Regulation is that interconnection with a major supplier should be:

- A. Arranged privately between the largest carriers, on terms they keep confidential
- B. Offered at whatever price the major supplier judges the market will bear
- C. Provided on non-discriminatory terms, at cost-oriented rates and at reasonable request ✓**
- D. Reserved for operators that the incumbent selects as suitable partners

Answer: C. Provided on non-discriminatory terms, at cost-oriented rates and at reasonable request

50. On the question of regulatory independence, the WTO Reference Paper and the course material indicate that a regulator should be independent of operators, while independence from government is:

- A. Required by law, so the regulator must sit inside the sponsoring ministry
- B. Required by law, so the regulator must be formally separate from government in all respects
- C. Arranged so that the regulated operators jointly own and direct the regulator
- D. Not strictly required, though a greater degree of it is generally regarded as desirable ✓**

Answer: D. Not strictly required, though a greater degree of it is generally regarded as desirable

51. Within the framework for regulation, the laws that governments pass to enable regulation characteristically:

- A. Guarantee the incumbent a lasting and protected monopoly position
- B. Establish regulatory bodies and grant them powers to act ✓**
- C. Strip regulators of their duties to report and account for decisions
- D. Set the retail price of each regulated service across the country

Answer: B. Establish regulatory bodies and grant them powers to act

52. Which option best lists the categories of law that bear specifically on the telecommunications services industry?

- A. Maritime law, constitutional law and the criminal law of the state
- B. General tax law, ordinary employment law and the intellectual-property law of everyday commercial dealings
- C. Sector-specific telecom law, competition and fair-trading law, and other relevant law such as privacy ✓**
- D. Banking law, insurance law and environmental law from other sectors

Answer: C. Sector-specific telecom law, competition and fair-trading law, and other relevant law such as privacy

53. Measures agreed by a regional body such as ECOWAS, or by the WTO, take legal effect within a member or signatory country once they are:

- A. Transposed into national law in line with the treaty obligations ✓**
- B. Posted on the regional body's official public website

- C. Endorsed by a majority vote of the country's licensed operators
- D. Agreed at the regional level, taking effect at that same moment

Answer: A. Transposed into national law in line with the treaty obligations

54. The framework notes that regulatory functions may be split between two or more bodies. A risk a country must manage when it does so is that:

- A. A single national regulator will end up capturing the whole market for itself
- B. Operators will find it much harder to obtain a licence from the divided bodies
- C. A poorly designed split of functions can breed disputes and slow the making of decisions ✓**
- D. Spectrum will end up handed to operators free of charge under the divided regulatory arrangement

Answer: C. A poorly designed split of functions can breed disputes and slow the making of decisions

55. Ranked from the least to the most intrusive, the recognised styles of regulation run:

- A. Formal regulation, co-regulation, self-regulation, zero regulation
- B. Zero regulation, self-regulation, co-regulation, formal regulation ✓**
- C. Self-regulation, zero regulation, formal regulation, co-regulation
- D. Co-regulation, formal regulation, zero regulation, self-regulation

Answer: B. Zero regulation, self-regulation, co-regulation, formal regulation

56. An industry said to operate under zero regulation is one in which:

- A. A regulator supervises the firm's daily operations in fine detail
- B. The firms jointly set binding rules under an industry watchdog
- C. Market forces are left to work without outside intervention ✓**
- D. A government ministry sets the prices at which firms must sell

Answer: C. Market forces are left to work without outside intervention

57. Self-regulation, in which the players in an industry set and police their own rules, tends to work best when:

- A. The stakeholders share a strong convergence of interest ✓**
- B. The players are locked in sharp commercial conflict with one another
- C. One firm dominates the sector and mistrusts its smaller rivals
- D. A regulator steps into the industry's routine operating decisions

Answer: A. The stakeholders share a strong convergence of interest

58. A commonly cited advantage of self-regulation is that:

- A. It lifts the administrative cost of oversight from consumers and the industry alike
- B. It keeps firms from behaving opportunistically toward their competitors
- C. It removes the need for underlying legislation or a licensing regime
- D. Firms usually understand their own business better than an outside regulator would ✓**

Answer: D. Firms usually understand their own business better than an outside regulator would

59. Co-regulation is best described as an arrangement in which:

- A. The regulator works alongside industry players in a joint forum rather than ruling from outside ✓**
- B. Firms run their own affairs with little external oversight of their conduct
- C. A distant regulator dictates the firms' technical and pricing choices from outside the industry forum
- D. A foreign body takes over the domestic regulatory authority for the sector

Answer: A. The regulator works alongside industry players in a joint forum rather than ruling from outside

60. Although it can be very effective, co-regulation is particularly exposed to:

- A. Role ambiguity, where the players hold differing views of who will do what ✓**
- B. A sudden and wholesale loss of the regulator's formal legal powers overnight
- C. The capture of the regulator by organised consumer groups
- D. A blanket legal ban on collaboration of this kind between firms

Answer: A. Role ambiguity, where the players hold differing views of who will do what

61. Regulation is described as intrusive when the regulator:

- A. Sets out broad principles and leaves the methods to the firms themselves
- B. Steps back from the market and lets market forces work unaided
- C. Intervenes frequently and in fine detail in a company's day-to-day operations ✓**
- D. Confines itself to issuing occasional, non-binding guidance

Answer: C. Intervenes frequently and in fine detail in a company's day-to-day operations

62. A reason intrusive regulation is generally viewed as undesirable is that:

- A. It sharply lowers the total cost of supervising the whole industry closely
- B. It assumes competition will deliver good outcomes without any oversight
- C. It shifts responsibility for poor market outcomes from the regulator across onto the operating firms
- D. The regulator drifts into acting as a proxy manager of the industry, weakening operators' freedom ✓**

Answer: D. The regulator drifts into acting as a proxy manager of the industry, weakening operators' freedom

63. Good practice suggests regulation should be pervasive rather than intrusive, meaning it should:

- A. Shape operators' behaviour broadly while letting them run their own affairs within the rules ✓**
- B. Prescribe the exact technical solution each operator is required to adopt
- C. Have as little day-to-day effect on companies' operations as can be managed
- D. Apply narrowly to just one dominant operator while leaving the other firms in the market untouched

Answer: A. Shape operators' behaviour broadly while letting them run their own affairs within the rules

64. Among the main instruments through which regulation is carried out are authorizations, rules, determinations, consents and:

- A. Share buy-backs
- B. Guidelines ✓**
- C. Dividend payments
- D. Corporate tax returns

Answer: B. Guidelines

65. A system of authorization may grant specific authorizations and general authorizations, which correspond respectively to:

- A. Spectrum assignments and national numbering plans
- B. Individual operator licences and class licences ✓**
- C. Price-cap orders and interconnection determinations
- D. Formal consents and published regulatory guidelines

Answer: B. Individual operator licences and class licences

66. Determinations are instruments of ongoing regulation whose function is essentially to:

- A. Cancel an operator's licence at once, with no route of appeal against the decision
- B. Overturn the underlying legislation suddenly and without prior notice
- C. Offer advice that carries no legal weight and binds no one
- D. Flesh out the rules, for example when price controls are reviewed periodically ✓**

Answer: D. Flesh out the rules, for example when price controls are reviewed periodically

67. Within the instruments of regulation, a consent is best understood as:

- A. A negative determination that releases an operator from meeting a particular requirement ✓**
- B. An order that places a fresh additional obligation on the particular operator concerned in the matter
- C. An informal note that carries no legal standing in practice
- D. A financial penalty imposed on an operator for breaching a licence term

Answer: A. A negative determination that releases an operator from meeting a particular

requirement

68. Guidelines differ from rules and determinations in that guidelines:

- A. Bind operators more tightly than the formal conditions attached to their own operating licences
- B. Carry no legal force in themselves and are usually written in plain, accessible language ✓**
- C. Are issued by the courts following a successful appeal by an operator
- D. Cancel a licence at the very moment the operator is found to breach them

Answer: B. Carry no legal force in themselves and are usually written in plain, accessible language

69. A key reason regulators consult widely before finalising their instruments is that consultation:

- A. Transfers the formal power of decision straight to the parties that are being consulted
- B. Relieves the regulator of its duty to give reasons for a decision
- C. Gives industry ownership of, and early sight of, decisions and improves their quality ✓**
- D. Discourages the consulted parties from appealing the final outcome

Answer: C. Gives industry ownership of, and early sight of, decisions and improves their quality

70. An appeal against a regulator's action may be brought on the grounds of law (illegality), procedure (misadministration) or:

- A. Profitability (loss of revenue)
- B. Logic (irrationality) ✓**
- C. Popularity (public disapproval)
- D. Timing (administrative delay)

Answer: B. Logic (irrationality)

71. Which of the following is a recognised enforcement power a regulator may use against an operator that breaks the rules?

- A. Seizing ownership of a competitor's shareholding
- B. Personally setting the salaries paid to each of the operator's own board of directors
- C. Barring the operator's customers from switching providers
- D. Imposing fines, issuing compliance orders, or in serious cases revoking the licence ✓**

Answer: D. Imposing fines, issuing compliance orders, or in serious cases revoking the licence

72. On the funding of national regulatory authorities, the typical arrangement is that:

- A. Operators are barred by law from contributing to the regulator's budget
- B. Government carries the primary responsibility but usually recovers the cost from industry through fees ✓**
- C. The regulator supports itself mainly on the fines it collects from ordinary telecom subscribers each year
- D. The budget comes chiefly from foreign donor agencies and aid grants

Answer: B. Government carries the primary responsibility but usually recovers the cost from industry through fees

73. The idea of regulatory hazard captures the concern that:

- A. Regulators face genuine physical danger to themselves when they inspect live network sites in the field
- B. Regulation strips most of the commercial risk out of the telecom market
- C. Regulatory power, if used unwisely, can distort markets and damage the industry it oversees ✓**
- D. Operators face no consequences for repeatedly breaching the rules

Answer: C. Regulatory power, if used unwisely, can distort markets and damage the industry it oversees

74. Regulatory capture is said to occur when:

- A. The regulator becomes gridlocked and stops issuing decisions for a long spell
- B. The regulator comes to identify with the industry it oversees and serves that industry's interests ✓**

- C. Individual operators are taken over and then run directly by the regulator acting as their outright owner
- D. Consumers step in and take charge of the regulator's day-to-day running

Answer: B. The regulator comes to identify with the industry it oversees and serves that industry's interests

75. The information asymmetry that regulators face arises mainly because:

- A. Regulators are forbidden by statute from gathering the operators' detailed cost and operating data
- B. Consumers deliberately hide their true preferences from the regulator
- C. Operators are obliged by statute to withhold their internal financial records

D. The firms being regulated usually know far more about their business than the regulator does ✓

Answer: D. The firms being regulated usually know far more about their business than the regulator does

76. Regulators grant licences to companies mainly in order to authorise them to:

A. Operate telecommunications networks and provide telecommunications services ✓

- B. Manufacture and export telecommunications equipment to buyers in other markets
- C. Set the country's monetary and fiscal policy for the wider economy
- D. Audit the financial accounts of competing operators each year

Answer: A. Operate telecommunications networks and provide telecommunications services

77. Beyond granting permission to operate, a telecommunications licence characteristically sets out:

- A. The exact retail prices that every other operator in the country must charge
- B. The internal staffing structure and the full organisation chart of the sector's own regulatory authority
- C. The facilities a company may deploy, the services it may offer, and how it deals with other operators ✓**
- D. The personal income-tax liabilities of the company's individual directors

Answer: C. The facilities a company may deploy, the services it may offer, and how it deals with other operators

78. For major licences that also govern access to scarce resources such as spectrum and numbers, operators are commonly required to pay:

- A. A fixed share of yearly profits handed to the competing operators
- B. An upfront fee when the licence is awarded, followed by annual payments ✓**
- C. No payment, since major licences are usually granted free of charge
- D. A one-off refundable deposit returned in full when the licence expires

Answer: B. An upfront fee when the licence is awarded, followed by annual payments

79. Historically, telecom licences were issued separately for different sub-markets such as fixed, mobile or paging services. The defining feature of that approach was that the licences:

- A. Were designed to be broadly technology- and service-neutral
- B. Applied identical terms across each and every market segment
- C. Were rarely tied to a specific technology or class of service
- D. Drew distinctions between particular technologies and services ✓**

Answer: D. Drew distinctions between particular technologies and services

80. The convergence of technologies, which lets different platforms deliver overlapping services, has led some regulators to shift toward:

- A. Technology- and service-neutral licensing ✓**
- B. Tighter technology-specific licensing for each platform
- C. The withdrawal of the licensing regime altogether
- D. Long, protected exclusive licences for the incumbents

Answer: A. Technology- and service-neutral licensing

81. A technology-neutral licence to provide a telephone service is one that:

- A. Obliges the operator to use one particular government-specified fixed access technology
- B. Prevents the operator from offering any voice service of its own
- C. Leaves the operator free to choose between a fixed network and a wireless technology ✓**
- D. Requires the operator to buy from a single approved equipment vendor

Answer: C. Leaves the operator free to choose between a fixed network and a wireless technology

82. The two main categories of licence awarded to telecom companies are:

- A. Individual operator licences and class licences (general authorizations) ✓**
- B. Spectrum-only licences and separate telephone-numbering licences
- C. Import licences and export licences for telecom equipment
- D. Short-term provisional licences and longer-term probationary trial licences

Answer: A. Individual operator licences and class licences (general authorizations)

83. Individual operator licences are used chiefly where:

- A. The market segment has ample room for many further new entrants
- B. The segment involves no scarce technical resources of any kind
- C. The particular market segment has ample room for many further new entrants to join and compete on equal terms
- D. Licences are limited in number, dominance is likely, or scarce resources such as spectrum must be assigned ✓**

Answer: D. Licences are limited in number, dominance is likely, or scarce resources such as spectrum must be assigned

84. Class licences (general authorizations) are characterised by the fact that they:

- A. Are tailored individually to a single, specifically named operator within the market segment concerned
- B. Apply the same terms of operation to all businesses in a segment, such as internet service providers ✓**
- C. Are reserved for the use of the former state incumbent operator
- D. Apply in the cases where radio spectrum has to be assigned

Answer: B. Apply the same terms of operation to all businesses in a segment, such as internet service providers

85. A current trend favours replacing individual operator licences with general authorizations where practical, because doing so:

- A. Maximises the one-off upfront revenue that the state collects from each licensed operator at the award stage
- B. Simplifies regulation, supports a level playing field, and adapts more readily to market change ✓**
- C. Locks in the incumbent's protected position for a long period
- D. Removes the need for an independent regulator in the sector

Answer: B. Simplifies regulation, supports a level playing field, and adapts more readily to market change

86. One licensing objective treats basic telecommunications as an essential public service. This is used to justify:

- A. Lifting regulatory oversight the moment the market is privatised
- B. Reserving service provision for a single state-owned monopoly for the long term
- C. Keeping some controls so that services are provided in the public interest ✓**
- D. Letting operators serve just the most profitable customers

Answer: C. Keeping some controls so that services are provided in the public interest

87. Network roll-out targets and coverage obligations are often written into licences chiefly to advance which licensing objective?

- A. The raising of one-off government revenue from spectrum auctions
- B. The shielding of the incumbent operator from new competition
- C. The harmonising of company accounting and reporting practices

D. The expansion of networks and services, and universal service ✓

Answer: D. The expansion of networks and services, and universal service

88. When a state-owned incumbent is privatized, a licence remains necessary because it:

- A. Transfers full legal ownership and operational control of the regulator across to the investor
- B. Assures the investor of lasting freedom from future regulation
- C. Lifts the operator's duty to serve less profitable customers

D. Sets out what the investor is buying and what the government expects the operator to do ✓

Answer: D. Sets out what the investor is buying and what the government expects the operator to do

89. Licensing can be used to shape the structure of a market most directly by:

- A. Fixing the wholesale price of each input used on the network
- B. Deciding how many operators are licensed to provide a given service ✓**
- C. Setting the wages that an operator pays its own employees
- D. Choosing the equipment vendors that operators are required to use

Answer: B. Deciding how many operators are licensed to provide a given service

90. Licence conditions intended to create a level playing field and to curb an incumbent's abuse of dominance are usually referred to as:

- A. Universal-service obligations for underserved regions
- B. Spectrum-refarming and reallocation provisions
- C. Tariff-rebalancing and price-adjustment schedules
- D. Anti-competitive safeguards, or fair-trading conditions ✓**

Answer: D. Anti-competitive safeguards, or fair-trading conditions

91. As a licensing objective, finite resources such as radio spectrum, numbers and rights of way should be allocated:

- A. In full to whichever operator has the longest trading history
- B. On a first-come basis with little regard to how efficiently they are used
- C. Between operators fairly, efficiently and in the public interest ✓**
- D. To the operators that happen to be owned by the state

Answer: C. Between operators fairly, efficiently and in the public interest

92. Licensing generates government revenue in two characteristic ways:

- A. Seizure of operator profits and the taking of their network assets by the state
- B. Personal income taxes levied directly on the regulator's own staff
- C. Yearly donations collected from international agencies and donors
- D. One-off revenue from auctions and continuing revenue from annual licence fees ✓**

Answer: D. One-off revenue from auctions and continuing revenue from annual licence fees

93. Consumer-protection conditions in telecom licences commonly deal with matters such as:

- A. Billing accuracy, complaint-handling arrangements and required services such as emergency calls ✓**
- B. The internal shareholding structure of the operator's parent holding company and its subsidiaries
- C. The personal credit ratings of the operator's individual directors
- D. The choice of external auditors used by the regulator itself

Answer: A. Billing accuracy, complaint-handling arrangements and required services such as emergency calls

94. By setting out clearly the rights and duties of the operator and the regulator, a licence advances the objective of:

- A. Regulatory certainty, which builds confidence and helps attract investment ✓**
- B. Spectrum refarming carried out across the competing operators
- C. The removal of the regulator's remaining discretion over its own day-to-day decisions
- D. An exactly equal share of the national market for each operator

Answer: A. Regulatory certainty, which builds confidence and helps attract investment

95. The international allocation of radio spectrum, designed to maximise its use while limiting cross-border interference, is coordinated by the:

- A. World Trade Organization (WTO)
- B. Economic Community of West African States (ECOWAS)
- C. International Telecommunication Union (ITU) ✓**
- D. Association of the operators within each country

Answer: C. International Telecommunication Union (ITU)

96. Although telephone numbers are in principle plentiful, they are treated as a scarce resource partly because:

- A. There is a small and strictly fixed stock of usable telephone-number digits in existence around the world
- B. People can recall only a limited string of digits, and numbers must be issued in meaningful blocks ✓**
- C. Each telephone number needs its own dedicated radio frequency to work
- D. A given number can be assigned to just one country across the world

Answer: B. People can recall only a limited string of digits, and numbers must be issued in meaningful blocks

97. Number portability is a mechanism by which:

- A. The subscriber effectively owns the number and can carry it when switching providers ✓**
- B. The operator keeps lasting ownership of the telephone numbers that it issues to its subscribers
- C. Numbers are auctioned off periodically to whichever operator bids the most
- D. Subscribers take a new number each time they move to a different provider

Answer: A. The subscriber effectively owns the number and can carry it when switching providers

98. A regulator may auction spectrum to the highest bidder or assign it at a low administered price. Assigning spectrum at a low price is typically intended to:

- A. Maximise the one-off revenue the government collects
- B. Set aside the band for the use of a single operator
- C. Lower prices for users or encourage the rollout of new services ✓**
- D. Remove the practical need to manage radio interference between competing users

Answer: C. Lower prices for users or encourage the rollout of new services

99. A primary reason spectrum is assigned through instruments such as licences, rather than used freely, is to:

- A. Increase the stock of digits available for telephone numbers nationally
- B. Let each operator broadcast on whichever radio frequency it happens to prefer at the time
- C. Ensure it is distributed sensibly and that users do not interfere with one another ✓**
- D. Remove the need for any coordination of frequencies across borders

Answer: C. Ensure it is distributed sensibly and that users do not interfere with one another

100. The core principle of interconnection is that:

- A. Each operator should deliberately keep its own subscribers walled off from the networks of rivals
- B. Everyone should be able to communicate with everyone else, whichever network they belong to ✓**
- C. Operators of a broadly similar size are the ones required to interconnect
- D. Interconnection is required between the largest international carriers

Answer: B. Everyone should be able to communicate with everyone else, whichever network they belong to